

A high-arity counterexample to Pixton completeness in Chow

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Abstract

We disprove the Chow and rational-cohomological forms of Pixton’s completeness conjecture for his original relation system. We construct a formal tautological class outside the original Pixton relation span whose image is zero in the Chow ring and in rational cohomology. The example has genus 10^{60} and $3\binom{10^{60}}{3}$ markings.

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1 Introduction

The tautological ring of the moduli space of stable curves is generated by cotangent and kappa classes, together with the classes obtained by forgetting markings and gluing nodes. Describing all relations among these natural classes is a central problem in the intersection theory of moduli spaces. Pixton proposed a uniform system of stable-graph relations and conjectured that it is complete. We construct a relation outside that system: an alternating product of corrected small diagonals that vanishes in rational Chow, and therefore in rational cohomology.

1.1 The relation problem

Mumford initiated the systematic study of natural classes in the Chow ring of the moduli space of curves [Mum, Sections 4–5]. Faber’s calculations and geometric constructions on powers of the universal curve led to a proposed description of the tautological ring of the smooth unpointed space $\mathcal{M}_g$, including its Gorenstein property [Fab, Conjectures 1–2]. Faber and Zagier later gave explicit hypergeometric formulas for relations among the kappa classes. Pandharipande–Pixton proved these relations in Chow using stable quotients and formulated their completeness on $\mathcal{M}_g$ [PP, Theorem 1 and Conjecture 2].

On the compactification, a presentation must also account for boundary strata and their intersections. Arbarello–Cornalba and Graber–Pandharipande developed the forgetful and boundary calculus [AC, Section 1], [GP03, Appendix A]; in genus zero, Keel gave a presentation of the entire Chow ring by boundary divisors [Keel, Section 4, Theorem 1]. Pixton’s 2012 graph formula extends the Faber–Zagier system to $\overline{\mathcal{M}}_{g,n}$ and proposes a complete set of relations [Pix, Sections 1–3, Conjecture 2]. Pandharipande–Pixton–Zvonkine proved the cohomological validity of this system using the 3-spin CohFT [PPZ]; Janda proved its Chow validity using equivariant Gromov–Witten theory of $\mathbb{P}^1$ [Jan]. These results establish relations, while completeness asks whether any further relations exist.

We fix the original relation space precisely. Let $S_{g,n}$ be the strata algebra of formal decorated stable graphs, with decoration degree truncated at the dimension of each vertex factor. Its realization is

$$q_{g,n}: S_{g,n} \longrightarrow A^*(\overline{\mathcal{M}}_{g,n}; \mathbb{Q}), \quad [\Gamma, \gamma] \longmapsto \xi_{\Gamma*} \gamma.$$

The gluing pushforward generators are unnormalized, as in [PPZ, Section 0.3]. Let $\mathcal{P}_{g,n}$ be the span obtained by placing one of Pixton’s graph-formula relations at a vertex of an outer stable graph, placing arbitrary strata classes at the other vertices, and gluing. Pixton’s Proposition 1 shows that this span is already an ideal, stable under the gluing and forgetting pushforwards and pullbacks [Pix]. Thus it includes these formal consequences from the outset. The smaller 0/1-marking seed system recovers the same span under the operations described in [PPZ, Section 3.5]. Section 2 gives the full recipe and all conventions.

Writing $R^*(\overline{\mathcal{M}}_{g,n}) = \text{im } q_{g,n}$, the Chow completeness question is whether

$$\ker q_{g,n} = \mathcal{P}_{g,n}$$

for every stable pair. The known validity theorem gives the inclusion $\mathcal{P}_{g,n} \subseteq \ker q_{g,n}$. Our example makes it strict.

1.2 The counterexample

The construction uses g colors and one three-label factor for each increasing triple of colors. Each factor is a product of two rational-tail divisor sums, corrected by a class supported on two-node rational bridges. We multiply these factors and alternate over relabelings that preserve the color of each slot. Definition 2.3 specifies the resulting formal class Y .

Theorem 1.1. *Set*

$$g = 10^{60}, \quad D_* = \binom{g}{3}, \quad n = 3D_*.$$

The explicitly defined class $Y \in S_{g,n}^{2D_}$ of Definition 2.3 satisfies*

$$q_{g,n}(Y) = 0, \quad Y \notin \mathcal{P}_{g,n}.$$

Consequently Pixton’s original relation span does not exhaust the tautological Chow relations: the equality $\ker q_{g,n} = \mathcal{P}_{g,n}$ fails for this stable pair.

The same class gives the rational-cohomological consequence. To specify that formulation, write $q_{g,n}^{\text{CH}} = q_{g,n}$ and define, in formal degree d ,

$$q_{g,n}^H = \text{cl} \circ q_{g,n}^{\text{CH}}: S_{g,n}^d \longrightarrow H^{2d}(\overline{\mathcal{M}}_{g,n}(\mathbb{C}); \mathbb{Q}),$$

where cl is the rational cycle-class map for the complex Deligne–Mumford stack. Its image is the cohomological tautological ring $RH^* = \text{cl}(R^*)$. Completeness in both R^* and RH^* is explicitly recorded as Pixton’s conjecture in [Can, Section 1, Conjecture 1].

Corollary 1.2 (Rational-cohomological incompleteness). *For the stable pair and class Y of Theorem 1.1,*

$$q_{g,n}^H(Y) = 0 \quad \text{in } H^{4D*}(\overline{\mathcal{M}}_{g,n}(\mathbb{C}); \mathbb{Q}), \quad Y \notin \mathcal{P}_{g,n}.$$

In particular, $\mathcal{P}_{g,n} \subsetneq \ker q_{g,n}^H$.

Proof. Cycle class commutes with Chern classes, products, and proper pushforward. In particular, $\text{cl}(\xi_{\Gamma*}\gamma) = \xi_{\Gamma*}\text{cl}(\gamma)$, and the same compatibility applies to the forgetting-pushforward definition of kappa classes. Thus $q_{g,n}^H$ is the cohomological graph realization of [PPZ, Section 0.3], with the same vertexwise truncation and unnormalized generators. Applying cycle class to Theorem 1.1 gives $q_{g,n}^H(Y) = 0$; nonmembership remains an assertion in the same formal strata algebra. The strict inclusion follows from

$$\mathcal{P}_{g,n} \subseteq \ker q_{g,n}^{\text{CH}} \subseteq \ker q_{g,n}^H.$$

Passing from Chow vanishing to cohomological vanishing uses the second inclusion; it requires no injectivity of cycle class. $\square$

The theorem and corollary give negative answers to the Chow and rational-cohomological completeness questions for the full original system. The large parameters leave ample room in the support and color estimates; they are not proposed as minimal parameters.

Completeness is distinct from the Gorenstein property. The latter asks for nondegenerate multiplication pairings in complementary degrees, whereas completeness asks for all relations among the generators. Pixton proved that $R^*(\mathcal{M}_{25})$ is not Gorenstein while establishing Faber–Zagier completeness there [Pix26, Theorem 1.1 and Section 2.2]. Canning–Larson–Schmitt proved completeness together with non-Gorenstein behavior on the compact-type spaces $\mathcal{M}_6^{\text{ct}}$, $\mathcal{M}_{5,2}^{\text{ct}}$, and $\mathcal{M}_7^{\text{ct}}$ [CLS, Theorems 1 and 7]. Our relation lies on the full stable compactification.

There is also a distinction between completeness and comparison of methods for producing relations. Janda’s comparison theorem identifies the systems arising from cancellation of reconstruction singularities for convergent semisimple CohFTs with nonempty discriminant [JanD, Definition 3.3.1 and Theorem 3.3.6]. It concerns the relations generated by those reconstructions; it does not identify the entire Chow relation kernel. The comparison throughout this paper is with the explicitly fixed span $\mathcal{P}_{g,n}$.

1.3 Proof and methods

The proof compares two algebras of cubic coefficients. A formal super Frobenius algebra can carry independent odd cubic coefficients while satisfying the identities needed to annihilate the original Pixton span. The actual Chow correspondences of a curve satisfy an additional nilpotence identity obtained from a surface Quot scheme. The first algebra proves nonmembership; the second proves vanishing. Figure 1 separates these two assertions about Y .

The formal test. Section 3 constructs a linear map on decorated graphs by discarding graphs outside the rational-tails locus and evaluating rational branches by Frobenius multiplication and coproduct. This uses the graph calculus familiar from two-dimensional topological field theory [Abr, Section 2 and Theorem 3]. The odd symplectic curve pairing and the vanishing of contractions between isotropically supported corrections have precedents in Pandharipande–Zvonkine’s CohFT

constructions [PZ, Sections 1.1–1.5 and Remark 11]. Here the coefficient ring is the exterior algebra on independent θ_{ijk} , one for every color triple; we verify the graph identities directly. Every element of $\mathcal{P}_{g,n}$ has zero test, while the test of Y has a nonzero coefficient proportional to $\prod_{i < j < k} \theta_{ijk}$.

The smooth-family Chow identity. Section 4 splits the correspondence object of a smooth curve into two even summands and an odd symplectic summand. The coefficient-algebra construction is a concrete form of O’Sullivan’s semisimple tensor reconstruction [OS, Lemma 2.3.1]. It retains all actual relative Chow correspondences, so its faithfulness is an assertion about Chow, not about cohomological realization. The small diagonal projected to three odd summands gives an alternating cubic φ with odd coefficients. Over a pointed curve this is the modified diagonal of Gross–Schoen [GS, Equation (0.1) and Sections 2–3]; the family construction also retains the necessary base corrections.

Sections 5–6 prove the additional nilpotence. The surface is the square of the curve. A relative Quot scheme with fixed determinant produces a relation for products of the cubic on isotropic vector sets. Its obstruction theory uses derived perfect-complex moduli and determinant maps [TV, STV], virtual pullbacks [BF, Man], and virtual localization [GP]. Surface Quot localization and relative Chow identities appear in Marian–Oprea–Pandharipande [MOP1]; the point obstruction factors have the antecedent in [OP, Lemma 34 and Section 5.2]. Here the calculation keeps the determinant constraint, arbitrary irregularity, and every marked surface output.

The nested-Hilbert recursion of Ellingsrud–Göttsche–Lehn [EGL] reduces that marked calculation to decorated equality diagonals. Kong develops relative forms of this recursion and, for curve families, formulas retaining one and two marked outputs [Kong, Section 2.1 and Lemma 3.4]. We prove the version retaining an arbitrary finite set of surface outputs. Only bare triple blocks survive the relevant codimension bound. Section 6 computes their common scalar and proves it is nonzero, using symmetric-power and residue methods related to [MOP2, Equation (9), Theorem 2, and Lemma 3]. It then converts the isotropic identity into vanishing tests for sufficiently long products with arbitrary color inputs (Lemma 6.4). Finite faithful polarization returns a Chow identity modulo classes supported over the prescribed positive codimension in the base.

Passage to the compactification. Section 7 compares the corrected stable cubic with these projected smooth-curve cubics. The framework uses Hassett’s weighted stable curves [Has] and configuration-space blowups [FM, Rou]; Petersen’s proof of Tavakol’s presentation conjecture treats the corresponding rational-tails comparison over a smooth core [Tav, PetW]. The nodal comparison here requires a rational-bridge correction with coefficient $-1/2$. A three-position calculation shows that it cancels the remaining triple-exceptional contribution. The errors are then either a two-slot operation times a one-slot operation from a bounded span, or operations supported over positive base codimension.

Section 8 uses these errors to lower the dimension of the image of the supporting cycle in the base. Chow-localization support methods occur in Graber–Vakil’s filtration by rational components [GV, Theorem 1.1 and Section 4.2]; the induction here acts on arbitrary cycle multipliers over the base parametrizing the curve components. When one component carries almost all the genus, the surface identity advances support. Otherwise a repeated-color identity supplies a smaller vanishing factor, using the curve algebra centered at a marked node branch, Schur–Weyl alternation, and symplectic contraction. The final color budget leaves enough entries for every advancement, until the support is empty. Pullback from the weighted space then proves $q_{g,n}(Y) = 0$.

Thus Sections 2–3 construct and detect the formal candidate; Sections 4–6 prove the smooth-family identity; and Sections 7–8 extend its vanishing to the compactification. Table 1 collects the

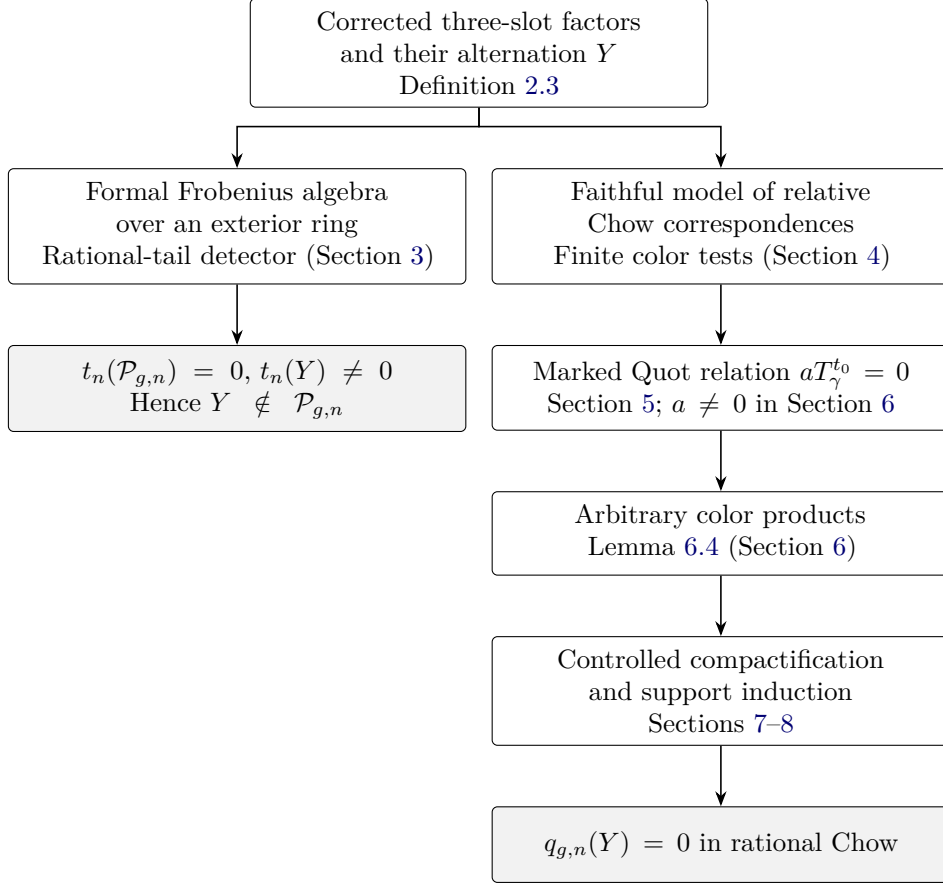


Figure 1: The two assertions about the same class Y . The formal branch is independent of Chow vanishing. In the geometric branch, the nonzero scalar completes the isotropic surface bound. The exterior-ideal argument then gives arbitrary-color vanishing modulo base support before the passage to the compactification.

notation used across these parts.

1.4 Conventions

All varieties, stacks, and families are over $\mathbb{C}$, and Chow groups have rational coefficients. On a smooth space we identify Chow classes with operational classes when pulling them to a test space. Products of stacks are understood in rational Chow theory [Kre, Theorem 2.1.12]. We use proper rational pushforward for relative Deligne–Mumford maps and its compatibilities from [BSch, Theorem B.17 and Propositions B.18–B.19]. Smooth Deligne–Mumford stacks have the operational identification in [BSch, Theorem C.6 and Corollary C.9]; for smooth quotient stacks, see also [EG, Propositions 4 and 19]. Every formal series calculation is truncated at a specified coefficient, so only finitely many terms occur.

Parity is distinct from Chow codimension. In a super tensor product, interchanging homogeneous factors of parities p, q gives $(-1)^{pq}$. In particular, on an odd vector space the super-symmetric tensors are ordinary alternating tensors. All finite averaging and polarization arguments take place in characteristic zero.

Notation	Meaning	Definition
$S_{g,n}, q_{g,n}, \mathcal{P}_{g,n}$	Formal strata algebra, Chow realization, original Pixton span	Section 2
D_*, n, Y	Number of cubic factors, number of labels, candidate relation	Definition 2.3
R, V, t_m	Exterior coefficient ring, formal Frobenius algebra, detector	Section 3
e'_C, η, H_C	Corrected degree-one class, pure projector, complementary curve object	Section 4
$\mathfrak{B}, W_b, \varphi$	Faithful Chow coefficients, odd symplectic space, projected cubic	Proposition 4.1
t_0, T_γ, a_{N,t_0}	Nilpotence exponent, isotropic cubic contraction, localization scalar	Sections 5–6
$F_{h,B}^0, G_{h,B}$	Corrected weighted cubic and its chosen pure lift	Section 7
$X^\Gamma, \mathcal{B}_\Gamma, \mathcal{S}_\Gamma$	Cut chart, compact heavy base, smooth-core open	Section 7

Table 1: Notation used across the proof. The formal coefficient ring R and the faithful Chow coefficient algebra $\mathfrak{B}$ have different roles.

2 The strata algebra and the explicit class

This section fixes the relation space. The distinction between an unnormalized gluing generator and an automorphism-weighted graph sum will be used in both parts of the proof.

2.1 Decorated graphs

A stable graph Γ of type (g, n) is a connected graph with n labeled legs and genera $g_v \geq 0$ at its vertices, satisfying

$$g = \sum_v g_v + |E(\Gamma)| - |V(\Gamma)| + 1, \quad 2g_v - 2 + n_v > 0.$$

Here n_v counts all incident half-edges, including legs. Its gluing map is

$$\xi_\Gamma: \prod_{v \in V(\Gamma)} \overline{\mathcal{M}}_{g_v, n_v} \longrightarrow \overline{\mathcal{M}}_{g, n}.$$

At every half-edge let ψ denote the cotangent class. At a vertex define

$$\kappa_j = \pi_*(\psi_{n_v+1}^{j+1}), \quad \kappa_0 = 2g_v - 2 + n_v,$$

where π forgets an additional marking.

The vector space $S_{g,n}$ is spanned by formal symbols $[\Gamma, \gamma]$, modulo graph isomorphism, with γ a monomial in the half-edge ψ 's and vertex κ_j 's for $j \geq 1$. At each vertex its decoration degree is at most $3g_v - 3 + n_v$. The total degree is $|E(\Gamma)| + \deg \gamma$, and

$$q_{g,n}[\Gamma, \gamma] = \xi_{\Gamma*}(\gamma).$$

There is no automorphism denominator in this definition of $q_{g,n}$. The per-vertex degree bound is the explicit convention in [PPZ, Section 0.3].

We use the usual decorated-graph intersection product. More explicitly, take common refinements with specified contractions to fixed representatives of the input graphs, requiring every refined edge

to survive in at least one contraction. Divide by the source symmetries preserving those contractions. For each additional use of an overlapping edge insert its normal factor $-\psi - \psi'$; pull the leg and flag decorations to their components and split kappa classes additively. Specified contractions record the lifts to the gluing parameter spaces, so this convention gives the product of the *unnormalized* pushforward generators. The construction is the transverse and self-intersection rule for node cuts [GP03, Appendix A, Proposition 9 and Equation (11)].

Terms exceeding any vertex dimension are set to zero. For multiplication and restriction to a further graph splitting, an excessive decoration remains excessive: the combined dimensions on the pieces decrease, so at least one piece still exceeds its dimension. Forgetting pullback of an undecorated graph is obtained by adding the new leg successively at each vertex. These conventions lift the corresponding Chow operations and do not impose any unproved relation. The preceding dimension argument is not an assertion about forgetting, which raises a vertex dimension. Compatibility of forgetting with the formal detector is checked separately in Section 3.

2.2 The exact Pixton span

Define

$$A(T) = \sum_{m \geq 0} \frac{(6m)!}{(3m)!(2m)!} T^m, \quad B(T) = \sum_{m \geq 0} \frac{6m+1}{6m-1} \frac{(6m)!}{(3m)!(2m)!} T^m.$$

For a commuting parity variable $\zeta^2 = 1$, put

$$\widehat{C}_{3j}(T, \zeta) = T^j A(\zeta T), \quad \widehat{C}_{3j+1}(T, \zeta) = \zeta T^j B(\zeta T).$$

Indices congruent to 2 modulo 3 are not used. For a series F write

$$\{F\} = \sum_{m \geq 0, \epsilon \in \{0,1\}} [F]_{T^m \zeta^\epsilon K_{m,\epsilon}} T^m.$$

Give every vertex of Γ its own variable ζ_v , with $\zeta_v^2 = 1$. Define the operator

$$\widehat{\kappa}_\Gamma \left(\prod_{i=1}^{\ell} K_{e_i, \epsilon_i} \right) = \sum_{\tau \in \mathfrak{S}_\ell} \prod_{c \text{ a cycle of } \tau} \left(\sum_{v \in V(\Gamma)} \kappa_{\sum_{i \in c} e_i}^{(v)} \zeta_v^{\sum_{i \in c} \epsilon_i} \right).$$

The empty product maps to 1, and the operator acts coefficientwise in T . The permutation-cycle expression is Faber's formula, as recorded in [AC, Equations (1.12)–(1.13)].

For an edge whose two half-edges lie at v, w , with cotangent symbols x, y , set

$$\Delta_e = \frac{A(\zeta_v x T) \zeta_w B(\zeta_w y T) + \zeta_v B(\zeta_v x T) A(\zeta_w y T) + \zeta_v + \zeta_w}{(x + y)T}.$$

This is a power series by

$$A(T)B(-T) + A(-T)B(T) + 2 = 0.$$

The same rule applies to a loop.

Definition 2.1 (Pixton's original span). Let $\sigma = (\sigma_1, \dots, \sigma_\ell)$ be a partition with no part congruent to 2 modulo 3, possibly empty, and let $a_1, \dots, a_n \geq 0$ also avoid that residue class. For $0 \leq r \leq 3g - 3 + n$, require

$$3r \geq g + 1 + |\sigma| + \sum_i a_i, \quad 3r \equiv g + 1 + |\sigma| + \sum_i a_i \pmod{2}. \quad (2.1)$$

Set

$$P_\Gamma = \left[2^{-h^1(\Gamma)} \widehat{\kappa}_\Gamma \left(\exp\{1 - \widehat{C}_0\} \prod_{j=1}^{\ell} \{\widehat{C}_{\sigma_j}\} \right) \cdot \prod_{i=1}^n \widehat{C}_{a_i}(\psi_i T, \zeta_{v(i)}) \prod_{e \in E(\Gamma)} \Delta_e \right]_{T^{r-|E(\Gamma)|} \prod_v \zeta_v^{g_v+1}}, \quad (2.2)$$

where $h^1(\Gamma) = |E| - |V| + 1$, $v(i)$ carries the i -th leg, and parity exponents are read modulo 2. In braces, the arguments of $\widehat{C}_i$ are T, ζ . A negative requested T -degree means zero. The relation is

$$\mathcal{R}(g, n, r; \sigma, a) = \sum_{\Gamma} \frac{[\Gamma, P_\Gamma]}{|\text{Aut } \Gamma|}.$$

The space $\mathcal{P}_{g,n} \subseteq S_{g,n}$ is the following linear span in each degree. Choose an outer stable graph of type (g, n) ; put one such relation at one vertex, and arbitrary decorated strata elements at all other vertices; then glue, using the unnormalized gluing operation. Take the span over every choice, including the graph with one vertex and no edges.

Proposition 2.2 (Known validity and stability). *The span in Definition 2.1 is an ideal and is stable under tautological gluing and forgetting pushforwards and pullbacks. For gluing pullback to a product, the target relation space is the sum of the spaces with a relation in one factor. Moreover*

$$\mathcal{P}_{g,n} \subseteq \ker q_{g,n}.$$

Proof. The ideal and operation statements are [Pix, Proposition 1] for precisely this original span. Chow validity of the $a_i \in \{0, 1\}$ seed relations follows from [Jan, Theorem 1 and its reconstruction]. The construction in [PPZ, Section 3.5] generates the higher marking indices by psi multiplication, the partition insertions by forgetting extra markings, and the outer-graph relations by gluing. These are formal identities, so they apply to the Chow-valid seeds. The parity condition in (2.1) converts the strict seed inequality to the displayed admissibility inequality. In Janda's reconstructed normalization, a degree- r seed is $(-1)^{\sum_i a_i} (-1/1728)^r$ times the empty-partition recipe above. This nonzero scalar does not change its vanishing or its span. $\square$

No converse to Proposition 2.2 is used. In particular, same-vertex multiplication of a relation does not enlarge the span, whereas an arbitrary new Chow relation is not assumed to belong to it.

2.3 The corrected cubic and its alternation

For the rest of the construction fix

$$g = 10^{60}, \quad D_* = \binom{g}{3}, \quad n = 3D_*. \quad (2.3)$$

Form a word $\mathbf{i}$ in the colors $1, \dots, g$ by concatenating all increasing triples of colors in lexicographic order. Its consecutive three-slot blocks will be denoted J . Each color occurs $\binom{g-1}{2}$ times.

For a set of labels T of size at least two, let D_T be the one-edge class cutting off a genus-zero tail with precisely those labels. For distinct labels a, b , put

$$\bar{D}_{ab} = \sum_{T \supseteq \{a, b\}} D_T.$$

We also need a class supported on rational bridges, not tails.

Definition 2.3 (The explicit counterexample). On three labeled slots $J = (a, b, c)$, take every one-edge *unmarked* stable graph of genus g , subdivide its edge by a genus-zero vertex, and put all three labels at that new vertex. Sum the resulting undecorated graphs, dividing each by its automorphism order. Pull this sum back by forgetting all labels outside J ; call the resulting class β_J^{st} .

Define

$$F_J^{\text{st}} = \bar{D}_{ab}\bar{D}_{ac} - \frac{1}{2}\beta_J^{\text{st}}, \quad Y = \sum_{\substack{\tau \in \mathfrak{S}_n \\ \tau i=1}} \text{sgn}(\tau) \tau \left(\prod_J F_J^{\text{st}} \right). \quad (2.4)$$

Here τ relabels slots, and products and forgetting pullbacks are the formal graph operations fixed above.

Every operation in Definition 2.3 is finite. Each cubic factor has codimension two, so

$$\deg Y = 2D_* < 3g - 3 + n.$$

The pair (g, n) is stable. The automorphism weights in β_J^{st} specify coefficients of this particular class; they do not change the unnormalized generators of $S_{g,n}$.

The proof of Theorem 1.1 now divides into two independent claims. Proposition 3.1 proves formal nonmembership. The remaining sections establish the geometric vanishing in Theorem 8.4.

3 A formal detector

We first separate the candidate from the relation space. Throughout this section the genus g is fixed as in Definition 2.3, but the number of labels is allowed to vary, including zero. No Chow relation on a positive-genus vertex will be imposed in the construction.

Proposition 3.1. *There are a supercommutative $\mathbb{Q}$ -algebra R , a finite free supermodule V over R , and $\mathbb{Q}$ -linear maps*

$$t_m : S_{g,m} \longrightarrow V^{\otimes_R m} \quad (m \geq 0)$$

such that $t_m(\mathcal{P}_{g,m}) = 0$ for every m , whereas $t_n(Y) \neq 0$. In particular $Y \notin \mathcal{P}_{g,n}$.

The maps are tests on formal strata. In particular, they are not asserted to factor through the Chow ring.

3.1 An isotropic cubic Frobenius algebra

The state space has the curve pairing familiar from Frobenius-algebra and CohFT constructions [Abr, PZ]. The exterior coefficient ring below allows an independent cubic on every triple of colors. We check associativity and the handle formula directly, since these particular identities govern the graph test.

Let

$$R = \Lambda_{\mathbb{Q}}(\theta_{ijk} : 1 \leq i < j < k \leq g),$$

with each generator odd. Let W be a purely odd symplectic vector space of dimension $2g$, with complementary Lagrangians W_+ and W_- . Choose bases $a_1, \dots, a_g$ and $b_1, \dots, b_g$ such that $\langle a_i, b_j \rangle = \delta_{ij}$. Extend the symbols θ_{ijk} alternately to all triples of indices. There is an alternating, odd-coefficient three-form Φ on W defined by

$$\Phi(a_i, a_j, a_k) = \theta_{ijk}, \quad \Phi(w_1, w_2, w_3) = 0 \quad \text{if some } w_i \in W_-.$$

Consider the free supermodule

$$V = R \otimes_{\mathbb{Q}} (\mathbb{Q}1 \oplus W \oplus \mathbb{Q}e),$$

where $1, e$ are even. Its trace, denoted by $\int$, takes e to 1 and is zero on $1, W$. Give V the even pairing which is hyperbolic on $\mathbb{Q}1 \oplus \mathbb{Q}e$, the prescribed symplectic pairing on W , and zero between these two summands. Declare 1 to be the unit, $e^2 = eW = 0$, and determine the remaining products by

$$\int w_1 w_2 = \langle w_1, w_2 \rangle, \quad \int w_1 w_2 w_3 = \Phi(w_1, w_2, w_3).$$

Equivalently, the product of two standard odd vectors is its symplectic pairing times e , plus the vector dual to the cubic functional in the third input. The latter vector has odd coefficients and belongs to $R \otimes W_-$.

Write μ for multiplication and Δ for its adjoint coproduct.

Lemma 3.2. *These rules give a supercommutative associative Frobenius algebra. The product of three standard odd vectors belongs to Re , and the product of four such vectors is zero. Its handle element is*

$$\mu\Delta(1) = (2 - 2g)e.$$

Proof. The pairing is nondegenerate, so the rules determine the product. Supercommutativity follows from alternation of the symplectic form and of Φ , with coefficients moved using the super sign rule. For example,

$$a_i a_j = - \sum_k \theta_{ijk} b_k, \quad a_i b_j = \delta_{ij} e,$$

and therefore $(a_i a_j) a_k = \theta_{ijk} e = a_i (a_j a_k)$.

More generally, pair the associator of three vectors from W with a fourth basis vector. Pairing with 1 gives the cyclic symmetry of the alternating cubic, and pairing with e gives zero. If the fourth vector is in W , both expressions are contractions of two pair-product tensors. Their W -channel contractions vanish because the cubic is supported on a Lagrangian; their hyperbolic-channel contractions vanish because $e^2 = eW = 0$. Nondegeneracy proves associativity. Cases involving 1 or e are immediate. The same product description proves the assertions about three and four odd inputs.

The hyperbolic channels contribute $2e$ to the handle element, and the $2g$ odd channels contribute $-2ge$. The cubic perturbation has zero internal trace, because a symplectic contraction pairs the two different Lagrangians. This gives the displayed formula. $\square$

Set

$$\lambda = (2g - 2)e.$$

For a nonempty finite label set L , write $\Delta_L : V \rightarrow V^{\otimes L}$ for the coproduct adjoint to iterated multiplication; for a singleton it is the identity. Coproducts are coassociative and satisfy the projection formula. In particular, writing $\Delta_{ij}(1)$ with units in all other slots, we have

$$\Delta_{ij}(1) \Delta_{ik}(1) = \Delta_{ijk}(1), \tag{3.1}$$

$$\Delta_{ij}(1)^2 = -\Delta_{ij}(\lambda). \tag{3.2}$$

$$\int_j \Delta_{ij}(v) = v. \tag{3.3}$$

The minus sign in (3.2) will match the normal bundle of a rational-tail boundary divisor.

3.2 Compression of rational tails

Call a graph a *rational-tails graph* if it is a tree with one genus- g vertex, called its root, and all other vertices rational. Every branch away from the root has nonempty labeled descendants. Such a graph has no automorphisms fixing the labels: the descendant label sets distinguish its branches, recursively.

We initially work without the degree cutoff, discard all graphs which are not rational-tails graphs, and allow genuine genus-zero Chow identities on rational parameter factors. This is only an auxiliary computation; we will prove that its output respects the cutoff in $S_{g,m}$. We use the genus-zero Chow boundary presentation of [Keel, Section 4, Theorem 1].

For each label i , put

$$\bar{\psi}_i = \psi_i - \sum_{\substack{T \ni i \\ |T| \geq 2}} D_T.$$

If the labels are ordered $1, \dots, m$, let L_x be the pullback of ψ_x from the space retaining just $1, \dots, x$. Thus

$$L_x = \psi_x - \sum_{\substack{T \ni x, |T| \geq 2 \\ T \setminus \{x\} \subseteq \{x+1, \dots, m\}}} D_T.$$

For $j \geq 1$, define

$$\bar{\kappa}_j = \kappa_j - \sum_{x=1}^m L_x^j.$$

The leading, graphless change of variables is $\bar{\psi}_i = \psi_i$ and $\bar{\kappa}_j = \kappa_j - \sum_i \psi_i^j$, which is an invertible polynomial change. Hence the bar monomials form a complement to the boundary subspace before truncation.

Define t_m on that complement by

$$t_m \left(\prod_i \bar{\psi}_i^{d_i} \prod_{j \geq 1} \bar{\kappa}_j^{c_j} \right) = \begin{cases} \lambda_1^{d_1} \cdots \lambda_m^{d_m}, & \text{if every } c_j = 0, \\ 0, & \text{otherwise.} \end{cases} \quad (3.4)$$

Here the tensor on the right has a unit in every unused slot. For a nontrivial rational-tails graph, integrate each non-root decoration on its genus-zero parameter space, retaining only its top-degree integral. Apply t_a to the decorated root with its a incident slots. Finally duplicate each root slot into the descendant labels of its branch by Δ_L . Every branch attached to the root has at least two descendant labels, so $a < m$; this is a recursive definition. For $m = 0$, the empty tensor is R , the unit maps to 1, and positive bar-kappa monomials map to zero.

Lemma 3.3. *The prescription is well defined modulo genuine relations on rational parameter factors and induces a map on $S_{g,m}$. For a homogeneous expression of degree r , its image is a sum of tensors indexed by partitions of the labels into nonempty blocks, of the form*

$$\bigotimes_L \Delta_L(\lambda^{d_L}), \quad r = \sum_L (|L| - 1 + d_L). \quad (3.5)$$

In particular t_m is zero in degrees $r > m$.

Proof. Grafting a genus-zero graph amounts to composing coproducts and multiplying the corresponding integrals. Coassociativity shows that the result is independent of the internal decomposition of a rational parameter class. In particular a zero class on such a factor contributes zero.

For a rational subtree with k descendant labels, top decoration degrees together with its edges, including the edge attaching it to the root, add to $k - 1$. Formula (3.5) follows by induction, starting from the bar prescription. Since $\lambda^2 = 0$, a nonzero term has $d_L \leq 1$, and hence $r \leq m$.

An excessive decoration on a rational vertex has zero integral. At a root with a incident slots, a decoration exceeding $3g - 3 + a$ also exceeds a , and its recursive test is zero. This applies since $g \geq 2$. Finally, under refinement the sum of the dimensions of the pieces of a vertex decreases by the number of new edges. Splitting a decoration additively between those pieces cannot make an excessive total degree admissible on every piece. Thus discarded terms also remain invisible under multiplication and refinement. Forgetting operations require a separate argument, supplied below. $\square$

3.3 Boundary operations and the test

We first show how the test treats multiplication by bar classes and rational-tail divisors. We then prove the forgetting identities that will reduce the number of labels in the relation argument. All tensor products, contractions, and permutations below use the super sign convention.

Lemma 3.4 (Boundary operations). *The maps t_m are equivariant for relabeling. For every strata expression b , they satisfy*

$$t(\bar{\psi}_i b) = \lambda_i t(b), \quad t(\bar{\kappa}_j b) = 0, \quad (3.6)$$

$$t(\bar{D}_{ij} b) = \Delta_{ij}(1) t(b). \quad (3.7)$$

More generally, gluing along D_T integrates the entire genus-zero input and duplicates the root-side attaching slot into the labels of T .

Proof. We first establish the boundary restrictions and the graph compatibilities used in the calculation. The descendant and kappa identities are the forgetting and gluing formulas in [AC, Equations (1.7)–(1.10)]. At the gluing for D_T , let p be the root-side flag and p' the rational-side flag. The normal class is $-\psi_p - \psi_{p'}$. On $\overline{\mathcal{M}}_{0,T \cup \{p'\}}$, the sum of boundary splits separating i from p' equals $\psi_i + \psi_{p'}$. The sum whose side away from p' contains two fixed labels i, j equals $\psi_{p'}$. These are the genus-zero cotangent boundary formulas; they also follow by successively forgetting to three labels and applying the cotangent pullback formula.

Consequently, $\bar{\psi}_i$ restricts to the modified cotangent of its ancestor slot on the root side, using p if $i \in T$. The restriction of $\bar{D}_{ij}$ is the analogous pair class on the root side, except when both labels lie in T , when it is $-\bar{\psi}_p$.

The class $\bar{\kappa}_j$ is independent of the chosen ordering. Indeed, for two consecutive labels made last, write $D = D_{\{i,j\}}$. The identities $\psi_i|_D = \psi_j|_D = 0$ and $D|_D = -\psi_p$ give, for $d \geq 1$,

$$(\psi_i - D)^d - \psi_i^d = -D\psi_p^{d-1} = (\psi_j - D)^d - \psi_j^d. \quad (3.8)$$

After pulling to the remaining labels, this is precisely the invariance under a consecutive transposition. To restrict $\bar{\kappa}_j$ to D_T , put the labels of T last. The kappa class splits additively between the two vertices. The first label of T gives the root-side cotangent at p ; the remaining partial cotangents on the rational factor cancel its kappa class by the forgetful kappa formula. The result is exactly $\bar{\kappa}_j$ on the root side.

These restrictions and the coproduct projection formula prove (3.6) recursively, beginning with (3.4). For (3.7), only the two-label tail has a nonzero undecorated integral. If i, j are in different branches, the root-side pair class joins their ancestor slots. If they are in the same branch, its

restriction is $-\bar{\psi}_p$, and the required identity is (3.2). This proves the assertion for an arbitrary boundary expression as well as for a bar monomial. The gluing assertion follows directly by coassociativity and the factorization of genus-zero integrals.

These calculations take place in the auxiliary formal graph space modulo relations on rational parameter factors. Simultaneous compatible splits produce the same tree independently of their order, with one normal factor for every repeated edge. Hence the polynomial multiplications used here commute and associate. A genuine zero on a rational factor remains zero: restriction there is a genuine genus-zero operation, and operations at another factor retain that zero with relabeled flags. Lemma 3.3 permits the dimension truncations. Graphs outside the rational-tails locus remain outside it under refinement, so their products contribute zero as well.

Finally, independence of the bar-kappa ordering, the restriction formulas, and the symmetric coproduct construction give equivariance under arbitrary relabeling. $\square$

3.4 Forgetting markings

The forgetting identities are needed in both directions: pullback will remove a zero-index leg from the Pixton recipe, and pushforward will test a tensor slot against the two even basis vectors.

Lemma 3.5 (Forgetting operations). *For every strata expression b , the maps t_m satisfy*

$$t(\pi_y^* b) = t(b) \otimes 1, \quad t(\pi_{y*} b) = \int_y t(b). \quad (3.9)$$

Here π_y forgets the label y ; the identities hold in any order of the remaining labels.

Proof. We first calculate before the degree cutoff, modulo genuine genus-zero relations, and check at the end that every discarded term has zero test after forgetting. No relation on a positive-genus parameter factor enters the calculation.

Formal pullback and projection. Forget pullback adds y at every possible vertex. At that vertex it replaces

$$\psi_h \longmapsto \psi_h - D_{\{h,y\}}, \quad \kappa_j \longmapsto \kappa_j - \psi_y^j. \quad (3.10)$$

A tail divisor pulls back as $D_T + D_{T \cup \{y\}}$. Against a tree, an incompatible split remains incompatible; a compatible new edge splits a vertex. If the split edge already occurs, the two pullback terms give its pulled normal class: adding y at an endpoint adds the boundary term cutting off that half-edge together with y . Thus pullback commutes with multiplication by the indicated polynomial classes.

The push-projection calculation is equally local. At a stable vertex, put

$$X_i = \pi_y^* \psi_i, \quad k_j = \pi_y^* \kappa_j, \quad d_i = D_{\{i,y\}},$$

so that $\psi_i = X_i + d_i$ and $\kappa_j = k_j + \psi_y^j$. The relevant rules are

$$d_i d_l = 0 \ (i \neq l), \quad d_i^2 = -X_i d_i, \quad \psi_y d_i = 0, \quad (3.11)$$

and

$$\pi_{y*} 1 = 0, \quad \pi_{y*} \psi_y^d = \kappa_{d-1} \ (d \geq 1), \quad \pi_{y*} d_i = 1. \quad (3.12)$$

They prove projection with graphless cotangent and kappa polynomials by substitution. For a tail multiplier, its two pullback splits distribute the κ_{d-1} in (3.12) additively over the two vertices; for d_i , exactly the compatible split pushes to the original tail. On a tree with a stable y -vertex this

reduces to the same vertex calculation. If forgetting contracts a three-point vertex with a surviving terminal label, the compatible split places y with that label. If it contracts a bridge, the two normal contributions add to the normal of the combined edge. All other compatible splits lift uniquely. This proves the needed push projection on boundary graphs as well.

Forgetting also preserves the rational-factor relations used in the auxiliary computation: its pullbacks and pushforwards on those factors are actual genus-zero operations. Operations at another factor retain a rational-factor zero with relabeled flags.

Pullback of the test. Both bar classes pull to themselves. On a boundary graph, adding y to a rational factor increases its dimension without increasing the pulled decoration degree, so its integral is zero. Adding y at the root uses the smaller recursive pull identity. This proves the pull assertion.

Pushforward of the test. For pushforward on a boundary graph, forgetting on a rational factor commutes with its integral; a three-point contraction uses the counit identity (3.3). Forgetting at the root again uses the smaller recursive assertion. On the bar-monomial complement, projection reduces the question to powers of $\bar{\psi}_y$, since all other bar classes are pulled back. The zeroth power pushes to zero. For $d > 0$, telescope

$$\psi_y^d - \bar{\psi}_y^d = \sum_{\emptyset \neq S} D_{S \cup \{y\}} \sum_{a=0}^{d-1} \psi_y^{d-1-a} \bar{\psi}_y^a. \quad (3.13)$$

On the rational tail with labels $S \cup \{y\}$, the dimension is $|S| - 1$, and only the term $a = d - |S|$ integrates. Its integral is $\int_{\overline{\mathcal{M}}_{0,|S|+2}} \psi_y^{|S|-1} = 1$. After pushforward and application of the test, (3.13) is therefore

$$\sum_{1 \leq |S| \leq d} \Delta_S(\lambda^{d-|S|}), \quad (3.14)$$

with units in all unused slots.

On the other hand,

$$t_m(\kappa_{d-1}) = \int \lambda^d + \sum_{1 \leq |S| \leq d} \Delta_S(\lambda^{d-|S|}). \quad (3.15)$$

For $d = 1$ this is the scalar identity $\kappa_0 = 2g - 2 + m$. For $d > 1$, expand $\kappa_{d-1} = \bar{\kappa}_{d-1} + \sum_x L_x^{d-1}$. Before pulling from the stage with labels $1, \dots, x$, write the exact polynomial identity

$$\psi_x^{d-1} - \bar{\psi}_x^{d-1} = \sum_{\substack{T \subseteq \{1, \dots, x\} \\ x \in T, |T| \geq 2}} D_T \sum_{a=0}^{d-2} \psi_x^{d-2-a} \bar{\psi}_x^a. \quad (3.16)$$

On the tail for T , the dimension is $|T| - 2$. Thus its integral selects exactly $a = d - |T|$, when that exponent lies between 0 and $d - 2$, and the integral of $\psi_x^{|T|-2}$ is 1. The remaining bar cotangent is the root-side cotangent, so this term contributes $\Delta_T(\lambda^{d-|T|})$, with coefficient $+1$. The term $\bar{\psi}_x^{d-1}$ gives the singleton $T = \{x\}$. Pulling through the later labels adds only unit tensor slots, by the pull identity already proved. Every block therefore occurs once, indexed by its largest label, with no binomial coefficient or normal self-intersection sign. Subtracting (3.14) from (3.15) proves $t_m(\pi_{y*} \bar{\psi}_y^d) = \int \lambda^d$, as required.

Compatibility with the cutoff. A non-rational-tails graph stays outside the test under forgetting: its nontrivial unmarked stabilization persists.

We must also check forgetting on terms discarded by the vertexwise cutoff; this does not follow from the refinement dimension argument alone. An excessive decoration on a rational factor is an actual zero in its genus-zero Chow ring. Its genuine forget pullback or pushforward is still zero, including the integral used when a three-point factor contracts. Operations on another factor preserve this rational-factor zero. For an excessive root decoration of degree d with a root slots, we have $d \geq 3g - 2 + a$. Adding a mark at that root gives an expression of degree $d > a + 1$, so the block bound kills its test with $a + 1$ slots. Forgetting a root mark gives degree $d - 1 > a - 1$, killed by the same bound with $a - 1$ slots. The root stays stable because $g \geq 2$. Adding or forgetting on a rational factor leaves the root decoration unchanged; a contracted rational three-point vertex merely replaces an attaching flag without changing the root arity. Thus discarded terms have zero test before and after each forget operation. Together with Lemma 3.3, this permits every cutoff in the displayed test identities. It does not assert that untruncated pullback descends unchanged to the truncated formal algebra. Equivariance from Lemma 3.4 allows the forgotten label to be placed last in any chosen order. $\square$

3.5 The zero-leg identity

We require one precise identity for the recipe in Definition 2.1. Package the partition insertions with formal variables s_ℓ , one for each allowed positive part, using exponential coefficients for repeated parts, and put

$$H(T, \zeta) = \widehat{C}_0(T, \zeta) - \sum_{\ell} s_{\ell} \widehat{C}_{\ell}(T, \zeta).$$

The variables s_{ℓ} are purely formal; all subsequent coefficient extractions involve finitely many of them. The logarithm below is expanded jointly in T and the s_{ℓ} , so its constant term at $T = s_{\ell} = 0$ is defined even though $H(0, \zeta)$ depends on the partition variables. The cycle exponential formula for $\widehat{\kappa}_{\Gamma}$ rewrites the vertex characteristic as

$$\exp\{-\log H\}_{\kappa, \zeta_v}. \quad (3.17)$$

Here a monomial $T^d \zeta^{\epsilon}$ in $-\log H$ is replaced by $\kappa_d T^d \zeta_v^{\epsilon}$, including $d = 0$. For the edges use $U(x, x') = T \Delta_e$, so the coefficient of total degree T^r is the degree- r recipe. Denote this packaged series, with its usual parity extractions, by $\mathcal{R}(s; a)$.

Lemma 3.6. *If all preceding leg indices a_j are zero or one, then, in the untruncated rational-tails computation,*

$$\mathcal{R}(s; a, 0) - \sum_{\ell} s_{\ell} \mathcal{R}(s; a, \ell) = \pi_y^* \mathcal{R}(s; a). \quad (3.18)$$

Proof. The kappa pullback formula, including $\kappa_0^{\text{old}} = \kappa_0^{\text{new}} - 1$, multiplies the new characteristic (3.17) by $H(\psi_y T, \zeta_v)$. This is exactly the linear combination of new leg factors on the left.

It remains to check bubbles introduced in pulling the old leg and half-edge cotangents. For a function $f(x)$, this pull gives the boundary correction $(f(0) - f(x))/x$, where x is the stable-side cotangent. At a new rational three-point vertex, its characteristic is $H(0, \epsilon)^{-1}$; the new H -leg cancels it. Let η, η' be stable-side parity variables, and write

$$A_x = A(\eta x T), \quad B_x = B(\eta x T), \quad A_{x'} = A(\eta' x' T), \quad B_{x'} = B(\eta' x' T).$$

The edge adjoining that bubble has factor

$$\frac{\eta(B_x + 1) + \epsilon(1 - A_x)}{x}. \quad (3.19)$$

If the bubble carries an old leg, its constant factor is 1 or $-\epsilon$. Extracting ϵ in (3.19) then gives $(1 - A_x)/x$ or $-\eta(B_x + 1)/x$, respectively. These are exactly the two divided differences.

If the bubble subdivides an old edge, the coefficient of ϵ in the product of its two edge factors is

$$\frac{\eta(B_x + 1)(1 - A_{x'}) + \eta'(B_{x'} + 1)(1 - A_x)}{xx'}. \quad (3.20)$$

Substituting the original edge numerator shows that (3.20) equals

$$\frac{U(0, x') - U(x, x')}{x} + \frac{U(x, 0) - U(x, x')}{x'}. \quad (3.21)$$

Thus it also gives precisely the two cotangent pullback corrections. Bubbles for distinct slots at a vertex are disjoint, and these are all possibilities for adding one mark. Finally, rational-tails graphs have $h^1 = 0$ and trivial automorphism groups, so neither the factor 2^{-h^1} nor unnormalized gluing introduces another coefficient. This proves the identity. $\square$

3.6 Annihilating the relation span

Lemma 3.7. *For every $m \geq 0$, $t_m(\mathcal{P}_{g,m}) = 0$.*

Proof. We induct first on arity. At a fixed arity, the induction already kills every nontrivial outer gluing and every relation multiplied by a tail divisor. With those terms disposed of, degree and partition length reduce the basic recipes to the all-ones case, where an alternating-form calculation finishes the argument.

Use the precise span and operation stability in Proposition 2.2, as established for the original relation family in [Pix, Proposition 1]. For a basic recipe relation, use the lexicographic order $(m, r, \ell(\sigma)) \in \mathbb{Z}_{\geq 0}^3$: first the number of labels, then degree, then the number of partition parts. At a fixed arity, all relation spaces at smaller arities, in every degree, have already been treated. At zero labels, every admissible basic relation has positive degree, whereas t_0 kills positive degrees.

Consider a generator obtained by a nontrivial outer gluing. If that outer graph is not a rational-tails graph, all its refinements have zero test. Otherwise, a relation on a rational factor has zero top integral, by its established Chow validity. A relation on the root is killed by the smaller-arity induction, because that root has fewer labels. Integration and coproducts then give zero for the entire generator.

We will also use the following consequence, valid for any $b \in \mathcal{P}_{g,m}$:

$$t_m(D_T b) = 0. \quad (3.22)$$

Indeed, write this product as the gluing pushforward of the gluing pullback of b . The latter belongs to the sum of the relation ideals in the two factors. Integrate the rational factor and use the smaller-arity induction on the root factor, as above. The root arity is $m - |T| + 1 < m$. This argument needs no induction on the degree of b .

It remains to consider a basic recipe relation. If $a_i = 3k + h$, with $k > 0$ and $h = 0, 1$, its defining leg factor shows that the relation is ψ_i^k times the degree- $(r - k)$ relation with $a_i = h$. The admissibility inequality and parity remain unchanged after subtracting $3k$ on both sides. Expand $\psi_i = \bar{\psi}_i + \sum_{T \ni i} D_T$. The term containing only $\bar{\psi}_i$ uses the lower triple $(m, r - k, \ell(\sigma))$ and is killed

by (3.6). Every other term is $D_T b'$ with $b' \in \mathcal{P}_{g,m}$, by the ideal property, and is killed by (3.22). That identity uses smaller arity only, regardless of the degree of the remaining multiplier. We may therefore suppose that all leg indices are zero or one. We may also suppose $r \leq m$, by Lemma 3.3.

If a leg has index zero, extract its degree and partition coefficient in Lemma 3.6. The pullback term uses $(m-1, r, \ell(\sigma))$ and is killed by (3.9). A remaining term moves a part ℓ to the zero leg and has order $(m, r, \ell(\sigma) - 1)$. If $\ell = 1$, this is already the required strict decrease. Every other allowed positive part is $\ell = 3k + h$ with $k > 0$ and $h = 0, 1$. Reducing that leg first gives a bar-multiplied relation of order $(m, r - k, \ell(\sigma) - 1)$, while its tail terms are again covered by (3.22). Thus every invocation is well founded. These are legitimate recipe relations: moving a part preserves its total weight, hence both admissibility conditions, and removing the zero leg does not change them. Moreover $r \leq m \leq 3g - 3 + (m - 1)$, since $g \geq 2$, so the pullback term fits the lower-arity dimension bound. The identity can be used before cutoff, because the test commutes with pullback and kills the excessive terms.

Only the all-ones case remains. Its recipe is symmetric under permutations of the labels, and the same is true of its test tensor. Contracting any one slot with 1 gives the test of a forget pushforward, which vanishes by operation stability and smaller arity. Contracting with e does also: multiply first by $\bar{\psi}_y/(2g-2)$, use the ideal property, and then forget that slot.

It suffices, therefore, to test all slots on standard vectors of W . In the block expansion (3.5), a singleton has zero pairing with an odd input, a positive power of λ kills odd inputs, and a block of at least four odd inputs has zero product. Only undecorated pair and triple blocks survive. If there are s pairs and u triples, then

$$m = 2s + 3u, \quad r = s + 2u, \quad s + 3u = 3r - m \geq g + 1 + |\sigma| > g. \quad (3.23)$$

The permutation symmetry of the tensor makes its pairing an ordinary alternating form on the odd input vectors. Each symplectic pair uses one dual coordinate of W_+ ; each cubic uses three such coordinates. Thus every alternated term in (3.23) uses more than g coordinates from the g -dimensional space $W_+^\vee$ and is zero. All standard input tests vanish, so nondegeneracy of the pairing proves that the tensor itself is zero. $\square$

3.7 The nonzero value on the candidate

We finish the proof of Proposition 3.1. Every graph in a bridge correction β_f^{st} has nontrivial unmarked stabilization and is not a rational-tails graph. Its refinements, pullbacks, and products remain outside the support of the test. By (3.7) and (3.1), the remaining part of each factor of Y tests as

$$\Delta_{ab}(1)\Delta_{ac}(1) = \Delta_{abc}(1).$$

Pair slot j with the first-Lagrangian vector a_{i_j} prescribed by the color word in Definition 2.3. A block with increasing colors $i < j < k$ gives θ_{ijk} . The product of the unsymmetrized factors therefore evaluates, up to one fixed super sign, to

$$\Theta = \prod_{1 \leq i < j < k \leq g} \theta_{ijk} \neq 0,$$

with the product ordered as the blocks.

Let G_1 be the stabilizer of the word. For $\tau \in G_1$, transporting its super permutation across the pairing of odd input slots contributes $\text{sgn}(\tau)$; the input word itself is unchanged. This cancels the explicit sign of that summand in Y . No reordering of the exterior generators is required in this

comparison: all summands are compared with the same ordered evaluation. Every color occurs $\binom{g-1}{2}$ times, and hence

$$\langle t_n(Y), a_{i_1} \otimes \cdots \otimes a_{i_n} \rangle = \pm \left(\binom{g-1}{2}! \right)^g \Theta \neq 0. \quad (3.24)$$

Together with Lemma 3.7, this proves the proposition.

4 Faithful tensor tests in relative Chow

The formal detector and the Chow-vanishing argument use different coefficient algebras. For the latter we must retain all Chow correspondences, including those not generated by the pairing. We construct such a tensor model in this section. Its coefficient algebra is generally large; the point is its faithfulness, not a presentation of its generators.

Let

$$\pi: C \longrightarrow S$$

be a smooth projective family of connected curves of genus b , with a section, over an irreducible smooth quasi-projective complex variety. Fix a rational divisor class $e_C \in A^1(C)$ of relative degree one. The class e_C need not be the class of the section. Set

$$a_C = \pi_*(e_C^2), \quad e'_C = e_C - \frac{1}{2}\pi^*a_C, \quad \eta = [\Delta_{C/S}] - e'_{C,1} - e'_{C,2}. \quad (4.1)$$

Here and below products of curve families are over S .

For an integer $a > 0$, we may work modulo classes that vanish after restriction to $S \setminus Z$, for some closed subset $Z \subset S$ of codimension at least a . This convention does not mean taking numerical equivalence, nor does it mean testing individual geometric fibers.

Proposition 4.1 (Faithful Chow tensor test). *In the setup above, either in relative rational Chow or in the indicated quotient by base support, there is a supercommutative rational algebra $\mathfrak{B}$ with the following properties.*

- (i) *The curve correspondence algebra has a faithful tensor model on*

$$\mathfrak{B} \otimes (\mathbb{Q}1 \oplus W_b \oplus \mathbb{Q}e),$$

where $1, e$ are even and W_b is an odd symplectic space of dimension $2b$. The classes $1_C, e'_C$ correspond to $1, e$, and η projects onto W_b . Integration gives the hyperbolic pairing $\langle 1, e \rangle = 1$, $\langle 1, 1 \rangle = \langle e, e \rangle = 0$ on the two even summands and the symplectic pairing on W_b . All actual Chow correspondences between curve powers, not only pairing diagrams, are included faithfully in this model.

- (ii) *The projected small diagonal is an alternating three-form*

$$\varphi: W_b^3 \longrightarrow \mathfrak{B}_{\text{odd}}.$$

Cup product, integration, tensor products, permutations, and contractions are computed in the model with the Koszul signs.

(iii) Let U be a finite-dimensional ordinary rational color space. A pure colored tensor in

$$\left(\eta^{\otimes k} A^*(C^k) \otimes U^{\otimes k} \otimes \text{sgn}_k \right)_{\mathfrak{S}_k} \quad (4.2)$$

is zero if and only if all its evaluations using the same ordinary linear map $z: U \rightarrow W_b$ in every slot are zero in $\mathfrak{B}$. Finitely many rational maps z suffice for each fixed U, k . The assertion holds in the base-support quotient as well.

In particular, a product of projected cubic entries is tested by products of $\varphi(z(i), z(j), z(k))$, with the fixed Koszul sign coming from the ordering of the factors. A vanishing established by these tests is a vanishing in Chow, with the stated base-support bound.

4.1 Correspondences and the curve projector

We use the rational correspondence category whose objects are smooth projective S -families and whose morphisms are

$$\text{Hom}(P, Q) = A^*(P \times_S Q),$$

with all codimensions allowed. Composition is pullback to the triple product, intersection, and pushforward. Tensor product is relative product. Equivalently, this is the usual correspondence calculus with Tate twists forgotten. We take finite direct sums and split idempotents. The diagonal identifies every generating object with its dual; the evaluation for the curve is its intersection pairing followed by π_* .

The morphisms that vanish away from a closed base subset of codimension at least a form a tensor ideal. Indeed, a finite union of such subsets has the same codimension bound, and restriction to its complement commutes with every operation just described. We can therefore first quotient by this ideal and then split idempotents. Morphisms between the original objects are unchanged by the latter completion. Denote either the unquotiented category or this completed quotient by $\mathcal{C}$. Its tensor unit is the base S . The endomorphisms of that unit are $A^*(S)$, or their base-support quotient; positive-degree base classes remain part of the correspondence calculus unless the chosen quotient kills them.

The projection formula gives

$$\pi_* 1_C = 0, \quad \pi_* e'_C = 1, \quad \pi_* ((e'_C)^2) = a_C - a_C = 0. \quad (4.3)$$

Thus $1_C, e'_C$ span a nondegenerate hyperbolic plane in the curve object. The two corresponding projectors sum to $e'_{C,1} + e'_{C,2}$, so their orthogonal complement has the self-transpose projector η . Write this complementary object as H_C .

The categorical trace of the curve identity is $\pi_* c_1(T_{C/S}) = 2 - 2b$, by diagonal self-intersection. Each of the two unit summands has trace one. Consequently

$$\dim_{\mathcal{C}} H_C = -2b. \quad (4.4)$$

The remaining relation needed to model H_C by an odd symplectic space holds already in relative Chow.

Lemma 4.2 (Relative pairing relation). *On C^{2b+2} , one has*

$$\sum_{\mathcal{M}} \prod_{\{i,j\} \in \mathcal{M}} \eta_{ij} = 0, \quad (4.5)$$

where $\mathcal{M}$ ranges over the perfect matchings of $\{1, \dots, 2b+2\}$.

Proof. Let $A \rightarrow S$ be the relative Jacobian. The section of C permits a universal degree-zero line bundle on $C \times_S A$, normalized along that section [BLR, Section 9.4, Proposition 4; Section 8.2, Proposition 4]. We use the Poincaré bundle and biduality for abelian schemes in [BLR, Section 8.4, Theorem 5]. If ℓ_1, ℓ_2 are its two first Chern classes on $C \times_S A \times_S A$, put

$$P = (\pi_{A \times A})_*(\ell_1 \ell_2) \in A^1(A \times_S A), \quad u = \frac{1}{2} \Delta_A^* P \in A^1(A).$$

Base twists of the universal line do not affect P , because its relative degree is zero. The class P is symmetric and biadditive. In particular

$$[m]^* u = m^2 u. \quad (4.6)$$

We first prove $u^{b+1} = 0$ in relative rational Chow.

The Fourier method originates in the work of Mukai and Beauville [Muk, Theorem 2.2], [Bea, Section 2, Proposition 3]. We give the argument with its relative base factor. For the normalized Poincaré line on $A \times_S \hat{A}$, write its first Chern class as c , and define on total Chow

$$\mathcal{F}(\alpha) = p_{\hat{A}*}(p_A^* \alpha \exp(c)).$$

All exponential expressions are finite in each Chow ring. Convolution with the analogous transform for $\hat{A}$ has kernel

$$(+)^* K, \quad K = p_{A*} \exp(c),$$

on $A \times_S A$. This follows by the biadditivity of the Poincaré class and flat base change.

The class K is supported on the zero section 0_A . To see this, apply Grothendieck–Riemann–Roch [BS, Section 7] to the Poincaré line under p_A . The relative tangent bundle of an abelian scheme is pulled back from the base, so its Todd factor is an invertible base class. The derived pushforward of the Poincaré line vanishes away from $0_A(S)$: on a geometric fiber, a nontrivial degree-zero line on an abelian variety has zero cohomology, and perfect direct image and derived base change give the assertion [Stacks, Tags 0B91 and 068V]. The fiber statement can also be seen from the flat unitary-character description of Pic^0 : a nontrivial character has no harmonic Dolbeault forms on a flat torus.

Chow localization now gives $K = 0_{A*} \delta$ for $\delta \in A^*(S)$. This coefficient is unique: if $\pi_A: A \rightarrow S$, then $\pi_{A*} 0_{A*} = \text{id}$, so $\delta = \pi_{A*} K$. Its degree-zero component is nonzero. It is the fiber integral

$$\int_{A_s \times \hat{A}_s} \frac{c^{2b}}{(2b)!},$$

which is nonzero because the Poincaré class is the full-rank evaluation pairing between the two first cohomology spaces. This computation determines only a scalar, not a Chow class by its realization. The positive-degree ideal of $A^*(S)$ is nilpotent, so δ is invertible. Flat base change along addition identifies $(+)^* 0_{A*} \delta$ with the graph of inversion carrying the pullback of δ . Convolution therefore gives inversion on A , times this invertible base class. In particular $\mathcal{F}$ is injective.

The same biadditivity and projection formula show

$$\mathcal{F} \circ [m]_* = [m]^* \circ \mathcal{F}.$$

Furthermore its component

$$\mathcal{F}_j(\alpha) = p_{\hat{A}*} \left(p_A^* \alpha \frac{c^j}{j!} \right)$$

has $[m]^*$ -eigenvalue m^j , where $j \geq 0$. On the other hand, $[m]$ is finite flat of degree m^{2b} , and (4.6) gives

$$[m]_* u^{b+1} = m^{-2} u^{b+1}.$$

For a fixed integer $m > 1$, the number m^{-2} is distinct from every eigenvalue m^j occurring in the finite sum $\mathcal{F}(u^{b+1})$. Applying the product of the corresponding linear polynomials in $[m]^*$ shows that this transform is zero. Injectivity proves the claimed vanishing.

Choose $N > 0$ clearing the denominators of e_C . The maps

$$a_N: C \longrightarrow A, \quad x \longmapsto [Nx - Ne_C]$$

are defined relatively. On an additional curve coordinate, the pullback of the universal degree-zero line has class $N[\Delta] - Ne_C$, up to a twist from the parameter curve. Integrating the product for two parameter positions gives

$$(a_N \times a_N)^* P = N^2([\Delta] - e_{C,1} - e_{C,2} + \pi^* a_C) = N^2 \eta. \quad (4.7)$$

The possible base twists contribute zero because the other line has relative degree zero.

Pull back $u^{b+1} = 0$ along

$$(x_1, \dots, x_{2b+2}) \longmapsto \sum_i t_i a_N(x_i),$$

where the t_i are independent integers. Biadditivity gives a polynomial identity in these integers. The coefficient of $t_1 \cdots t_{2b+2}$ is

$$(b+1)! N^{2b+2} \sum \prod_{\{i,j\} \in \mathcal{M}} \eta_{ij}.$$

Indeed, the self terms have a squared multiplier and cannot contribute to this coefficient. Rational polynomial interpolation and division by the displayed nonzero scalar prove (4.5). $\square$

4.2 Retaining arbitrary Chow correspondences

Let $\mathcal{R}_b = \text{Rep}^{\text{odd}}(\text{Sp}_{2b})$. This is the usual semisimple rational representation category of the split symplectic group, with parity prescribed by its central involution. Its standard representation W_b is odd. The ordinary alternating symplectic form is therefore a symmetric pairing in the super sense; its closed loop has value $-2b$.

The first and second fundamental theorems for symplectic tensors present this category, after finite direct sums and idempotent completion, by pairing diagrams with that loop value and the sum-of-matchings relation on $2b+2$ slots. In this formulation crossings on the standard odd space are minus the ordinary swap. See [RW, Theorems 2.2 and 2.13], or [LZ, Theorems 4.6, 4.8, and 5.9]. The diagram spaces and relations are rational, so the complex formulation descends by faithful scalar extension to $\mathbb{Q}$. Complete reducibility, absolute simplicity, and tensor generation over $\mathbb{Q}$ follow from [Mil, Theorems 4.14, 22.2, and 22.42, and Paragraph 22.4]. In particular, the irreducibles occur as rational direct summands of tensor powers of the standard representation. Equations (4.4) and (4.5) thus give a symmetric tensor functor

$$F: \mathcal{R}_b \longrightarrow \mathcal{C}, \quad F(W_b) = H_C. \quad (4.8)$$

For $b = 0$, (4.5) says $\eta = 0$, which gives the same conclusion with $W_0 = 0$.

It is essential not to assume that (4.8) is full. The target contains the small diagonal and many other correspondences. We record them by enlarging the coefficient algebra. The following reconstruction is a concrete split semisimple form of O'Sullivan's tensor realization [OS, Lemma 2.3.1(i)].

It applies to a symmetric tensor functor with this semisimple source and an arbitrary rational tensor target; the endomorphism ring of the target unit need not be a field. We give its coefficient construction explicitly.

Lemma 4.3 (Coefficient algebra). *Let F be the functor in (4.8), and put $H(X) = \text{Hom}_{\mathcal{C}}(1, F(X))$. There is a supercommutative Sp_{2b} -algebra $\mathfrak{B}$ such that, naturally in X ,*

$$H(X) \simeq (\mathfrak{B} \otimes X)^{\text{Sp}_{2b}}. \quad (4.9)$$

Under these identifications, tensor products of target morphisms become multiplication of coefficients. Arbitrary morphisms $F(X) \rightarrow F(Y)$ are represented faithfully by equivariant $\mathfrak{B}$ -linear maps $\mathfrak{B} \otimes X \rightarrow \mathfrak{B} \otimes Y$, and their composition is the usual tensor contraction.

Proof. Choose one representative X_λ of each irreducible rational symplectic representation and set

$$\mathfrak{B} = \bigoplus_{\lambda} H(X_\lambda) \otimes X_\lambda^\vee. \quad (4.10)$$

The multiplicity spaces $H(X_\lambda)$ are ordinary vector spaces; the parity on each summand comes from $X_\lambda^\vee$. Semisimplicity gives

$$H(X) \simeq \bigoplus_{\lambda} H(X_\lambda) \otimes \text{Hom}_{\text{Sp}_{2b}}(X_\lambda, X),$$

which is exactly (4.9). Only finitely many summands enter for any fixed X .

An element $s \in H(X)$ corresponds equivalently to an equivariant map $i_s: X^\vee \rightarrow \mathfrak{B}$. For $s \in H(X)$ and $t \in H(Y)$, the target tensor product gives

$$s \otimes t \in H(X \otimes Y).$$

Define multiplication on $\mathfrak{B}$ by requiring the product of i_s and i_t to be the map corresponding to this element, using the natural super identification $X^\vee \otimes Y^\vee \simeq (X \otimes Y)^\vee$. Explicitly, for homogeneous dual vectors f, g this identification sends $f \otimes g$ to

$$x \otimes y \mapsto (-1)^{|g||x|} f(x)g(y).$$

On the irreducible summands of (4.10) this specifies every product; naturality and semisimplicity give the same rule for all X, Y . Associativity, the unit, and supercommutativity follow from the associativity, unit, and symmetry constraints for the target tensor product. Thus (4.9) respects tensor products, including their Koszul signs.

Finally rigidity identifies

$$\text{Hom}_{\mathcal{C}}(F(X), F(Y)) \simeq H(X^\vee \otimes Y).$$

Apply (4.9) and the usual tensor–Hom identification for finite free supermodules:

$$H(X^\vee \otimes Y) \simeq (\mathfrak{B} \otimes X^\vee \otimes Y)^{\text{Sp}_{2b}} \simeq \text{Hom}_{\mathfrak{B}, \text{Sp}_{2b}}^{\text{even}}(\mathfrak{B} \otimes X, \mathfrak{B} \otimes Y).$$

This gives the asserted faithful representation of every target morphism. Evaluation and coevaluation are images of the corresponding representation morphisms, so composition is precisely coefficient multiplication and contraction. Faithfulness here is the injectivity of an actual vector-space isomorphism, not faithfulness of a cohomological realization. $\square$

Apply the lemma to

$$V_b = \mathbb{Q}1 \oplus W_b \oplus \mathbb{Q}e, \quad F(V_b) \simeq C.$$

In particular, a pure Chow class on C^k is a morphism $1 \rightarrow H_C^{\otimes k}$, so it is identified with an invariant in $\mathfrak{B} \otimes W_b^{\otimes k}$. This identification retains the class itself, including any component invisible to cohomology. The small diagonal and the structure map supply multiplication, unit, and integration. Their correspondence identities give a supercommutative Frobenius algebra on $\mathfrak{B} \otimes V_b$. The pairing is the one fixed by (4.3). In particular

$$\varphi(w_1, w_2, w_3) = \int_C w_1 w_2 w_3 \quad (w_i \in W_b) \quad (4.11)$$

is the tensor of the small diagonal after projection by η in all three slots. Its coefficients are odd, and it is alternating in the ordinary underlying vectors w_i . For a pointed curve over a field, this projection is the modified diagonal of Gross and Schoen [GS, Equation (0.1) and Sections 2–3]. The normalization here also incorporates the base corrections required for a family.

Lemma 4.4 (Centering at a section). *Work modulo positive-codimension base support, and suppose e_C is the class of a section. Then $e'_C = e_C$ in the quotient. In the model of Proposition 4.1, the functional*

$$\epsilon(x) = \int_C e_C x$$

is multiplicative and satisfies

$$\epsilon(1) = 1, \quad \epsilon(e) = 0, \quad \epsilon(W_b) = 0.$$

Consequently the component of a product $w_1 w_2$, with $w_1, w_2 \in W_b$, in the two even summands is $\langle w_1, w_2 \rangle e$.

Proof. Every positive-degree base class vanishes after removing a suitable positive-codimension subset. In particular a_C vanishes, so $e'_C = e_C$. Pairing with e_C is pullback along the section, by the projection formula, and this pullback is multiplicative. The displayed values follow from the hyperbolic splitting and the orthogonality of H_C .

The coefficient of 1 in $w_1 w_2$ is its pairing with e , namely $\epsilon(w_1 w_2) = 0$. The coefficient of e is its pairing with 1, namely $\langle w_1, w_2 \rangle$. These are identities of actual correspondences, hence also identities in the faithful coefficient model. $\square$

4.3 Colored operations and finite polarization

For an ordinary rational color space U , define

$$\mathcal{A}_k(U) = \left(A^*(C^k) \otimes U^{\otimes k} \otimes \text{sgn}_k \right)_{\mathfrak{S}_k}.$$

The product is pullback to disjoint curve slots followed by the coinvariant projection. With parity $k \bmod 2$ in arity k , the direct sum of these spaces is supercommutative: interchanging blocks of k and ℓ slots has sign $(-1)^{k\ell}$. The same definition applies to a collection of configuration spaces with equivariant forgetting maps, using their forgetting pullbacks for the product. We will use that version for weighted pointed curves.

A symmetric three-slot class gives an odd alternating entry $f(i, j, k)$, multilinear in its colors. For the pure small diagonal, its scalar-color evaluation is $\varphi(z(i), z(j), z(k))$. We use a fixed ordering of tensor factors throughout; reordering several entries introduces exactly their Koszul signs.

Lemma 4.5 (Finite color polarization). *In the pure subspace $\eta^{\otimes k} \mathcal{A}_k(U)$, evaluations using the same scalar map $z: U \rightarrow W_b$ in every slot are jointly injective. For fixed k, U , a finite collection of rational maps is jointly injective. These statements hold after quotienting by base support of any prescribed positive codimension.*

Proof. By Lemma 4.3, before taking color coinvariants a pure class is an invariant tensor with coefficients in $\mathfrak{B}$ and k odd standard factors. The super permutation of those factors is the ordinary permutation multiplied by sgn_k . It cancels the explicit sign representation in the definition of $\mathcal{A}_k(U)$. Writing $W_{b,\text{ord}}$ for the underlying ordinary vector space, the invariant-tensor inclusion therefore induces an injection

$$\eta^{\otimes k} \mathcal{A}_k(U) \hookrightarrow \mathfrak{B} \otimes \text{Sym}^k(W_{b,\text{ord}} \otimes U).$$

The symmetric group acts trivially on the single coefficient factor $\mathfrak{B}$ in this formula.

Using the nondegenerate symplectic pairing to identify a standard factor with its dual, the colored tensor is therefore an ordinary homogeneous polynomial tensor of degree k on $\text{Hom}_{\mathbb{Q}}(U, W_b)$, with coefficients in $\mathfrak{B}$. The invariant-tensor identification is injective, and taking finite symmetric-group coinvariants is exact over $\mathbb{Q}$. Hence no additional kernel was introduced in obtaining this polynomial tensor.

Evaluation at z is precisely polynomial evaluation. A homogeneous polynomial with coefficients in any rational vector space is determined by its values on rational points. More explicitly, after choosing bases, interpolation on a finite rational grid recovers every degree- k coefficient. The interpolation matrix has rational entries and a rational left inverse. It therefore works without requiring the coefficients to be reduced, even, or finite-dimensional. This proves both injectivity assertions.

The proof can be carried out from the start in the quotient correspondence category. Alternatively, use the same finite interpolation: take the union of the finitely many closed base supports occurring in the evaluations and coefficient identifications. Its codimension is at least the prescribed bound. Restriction to its complement makes every evaluation zero, and the interpolation identity makes the original colored class zero there. This explains why a separate exceptional support for each test does not weaken the conclusion. $\square$

Proof of Proposition 4.1. The hyperbolic decomposition, Lemma 4.2, and the symplectic diagram presentation construct the functor F . Lemma 4.3 supplies the faithful model for all curve correspondences and therefore for the Frobenius algebra. Equation (4.11) identifies the projected small diagonal. Lemma 4.5 supplies the same-map test and its base-support assertion. Tensor products represent products of disjoint-slot operations, so the final statement follows with their Koszul signs. $\square$

5 A surface relation from determinant-fixed Quot schemes

The tensor formalism of Section 4 does not by itself bound products of the odd coefficients of a projected small diagonal. We obtain the additional relation we need from the surface square of the curve. The decisive features are a determinant constraint, which removes the surface obstruction directions, and a marked Hilbert-scheme calculation, which retains the Chow classes of all output positions.

5.1 The surface bound

Let $\pi_C: C \rightarrow S$ be a smooth projective family of connected curves of genus b , with a section, over an irreducible smooth quasi-projective complex variety. Choose a rational divisor e_C of relative

degree one, and use the corrected divisor e'_C , projector η , and odd alternating cubic form φ of Proposition 4.1. Set

$$\chi = (b-1)^2, \quad s_K = 8\chi, \quad d = 3, \quad t_0 = \chi + \frac{d(d-1)}{2}s_K - 1 = 25\chi - 1, \quad m = 3t_0. \quad (5.1)$$

Throughout this section, the support quotient is the quotient by Chow classes that vanish after deleting a closed subset of S of codimension at least $b-1$. Its coefficient superalgebra will be denoted by $\mathfrak{B} = \mathfrak{B}_{S,b-1}$.

Theorem 5.1 (Surface bound). *Suppose $b \geq 5$ and $\dim S < \chi - 1$. Let $(w_\alpha)_{\alpha \in I}$ be any finite collection of vectors in the odd symplectic standard space W_b whose span is isotropic. For any matrix of scalars $\gamma = (\gamma_{\alpha\beta})_{\alpha,\beta \in I}$, put*

$$T_\gamma = \sum_{\substack{\alpha_1, \alpha_2, \alpha_3 \in I \\ \beta_1, \beta_2, \beta_3 \in I}} \left(\prod_{j=1}^3 \gamma_{\alpha_j \beta_j} \right) \varphi(w_{\alpha_1}, w_{\alpha_2}, w_{\alpha_3}) \varphi(w_{\beta_1}, w_{\beta_2}, w_{\beta_3}). \quad (5.2)$$

Then

$$T_\gamma^{t_0} = 0 \quad \text{in } \mathfrak{B}. \quad (5.3)$$

The assertion remains valid after any characteristic-zero scalar extension of the tensor formalism.

Isotropic tests alone do not separate arbitrary Chow tensors. In Lemma 6.4, we first fix an arbitrary color map, find a subspace of its domain with isotropic image, and use the exterior-ideal argument to annihilate the entire sufficiently long product, throughout modulo base support of codimension at least $b-1$. Only after doing this for every color map do we apply the faithful test of Proposition 4.1.

We prove the localization relation underlying this theorem in the present section. Section 6 proves that its scalar coefficient is nonzero and thereby completes the proof.

Write

$$X = C \times_S C, \quad \pi_X : X \longrightarrow S, \quad K = K_{X/S}.$$

We write line bundles additively. On every fiber, K is ample, $K^2 = s_K$, and $\chi(\mathcal{O}_X) = \chi$. The curve section gives a section $o : S \rightarrow X$. The two summands of the target bundle and their torus weights are

$$V = L_1 \oplus L_2 = K \oplus \mathcal{O}_X, \quad (w_1, w_2) = (0, 1), \quad A_0 = L_1 - L_2 = K. \quad (5.4)$$

Here the use of w_1, w_2 for weights is confined to formulas concerning the target bundle; the tensor vectors retain their indices α . Let $\mathcal{Q}$ be the relative Quot scheme of rank-zero quotients

$$0 \longrightarrow I' \longrightarrow V \longrightarrow \mathcal{Q} \longrightarrow 0$$

with the following determinant and numerical data:

$$D = dK, \quad \det \mathcal{Q} = D \text{ in } \text{Pic}(X/S), \quad \int_{X_s} \text{ch}_2(\mathcal{Q}_s) = \nu + L_1 D - \frac{D^2}{2}, \quad \nu = t_0 - 1. \quad (5.5)$$

All intersections in this display are fiber intersections. One may use the relative polarization K to specify the Hilbert polynomial. Projective representability and the Grassmannian construction are supplied by Grothendieck's Quot theorem [Nit, Theorem 5.1 and Section 5]. The determinant condition is a closed fiber of the separated relative Picard scheme [Kle, Theorem 4.8(1)], so $\mathcal{Q}$ is projective over S .

5.2 The derived obstruction theories

The determinant constraint changes both the obstruction space and the virtual dimension of $\mathcal{Q}$. We first construct its tangent complex as the fiber of the determinant differential. The surjectivity of that differential on obstruction spaces will give a two-term theory. The same construction on the torus-fixed factors will then express their virtual classes as virtual pullbacks from one rank-one Quot scheme.

5.2.1 Arrows and the Quot tangent

The perfect-complex moduli theorem of [TV, Corollaries 3.15, 3.17, and 3.19, Definition 3.18, Lemma 3.27, and Corollary 3.29] applies over every affine open of S . Indeed, the dg-category of perfect complexes on a smooth proper scheme over an affine base is proper by perfection of relative derived Hom and smooth by perfection of its diagonal. These smooth proper categories are saturated in the sense of [TV, Lemma 3.27 and Corollary 3.29]. Their moduli of perfect objects and perfect arrows are locally geometric and locally of finite presentation. These constructions glue over S . The tangent of the object moduli at a perfect complex F is $\mathbf{R} \operatorname{Hom}(F, F)[1]$. All Hom complexes in what follows are derived Hom on the surface, followed by derived pushforward to the parameter scheme. Perfect direct images and derived base change are used in the form of [Stacks, Tag 0B91], and relative duality in the form of [Stacks, Tags 0BRT, 0A9Q, and 0B6I].

Lemma 5.2 (The open arrow enhancement). *The derived stack of arrows with target V has an open derived substack whose classical truncation is the ordinary relative Quot scheme with the prescribed Hilbert polynomial, before fixing determinant. On this truncation its relative tangent complex is*

$$\operatorname{Cone}(\mathbf{R} \operatorname{Hom}(I', I') \longrightarrow \mathbf{R} \operatorname{Hom}(I', V)) \simeq \mathbf{R} \operatorname{Hom}(I', Q).$$

This is a perfect complex of Tor-amplitude $[0, 1]$, compatible with base change. The differential of forgetting the arrow and keeping its source is the boundary map

$$\mathbf{R} \operatorname{Hom}(I', Q) \longrightarrow \mathbf{R} \operatorname{Hom}(I', I')[1].$$

Proof. In the perfect-arrow stack take the locus where the source and cone are degree-zero sheaves on surface fibers, with the prescribed Hilbert data. For a perfect complex, relative Tor-amplitude zero is open. Concretely, a locally free resolution can be truncated first at its positive end; at its negative end, fiberwise injectivity and the local flatness criterion successively give flat cokernels. Properness of the surface family makes the condition open on the parameter scheme. Thus on a classical base the selected arrows are injections with flat sheaf quotients.

Conversely, the kernel and quotient of a flat quotient family are perfect on the total surface family, including over a singular parameter scheme. In a local free resolution the successive syzygies remain flat over the parameters. Regularity of the surface fibers makes the second syzygy fiberwise locally free; the local flatness criterion then makes it locally free on the total space. Hence the resolution terminates. These arguments commute with base change. An automorphism of the source over its inclusion in V is the identity, and negative Ext groups of the sheaves vanish. The classical moduli problem is therefore the Quot scheme, not a gerbe over it.

For an arrow $F \rightarrow G$ with both vertices variable, the endomorphism complex of the diagram is

$$\operatorname{fib}(\mathbf{R} \operatorname{Hom}(F, F) \oplus \mathbf{R} \operatorname{Hom}(G, G) \longrightarrow \mathbf{R} \operatorname{Hom}(F, G)),$$

where the map is the difference of pre- and postcomposition. Its shift by $[1]$ is the tangent of the arrow moduli. Taking the tangent fiber over the fixed target gives the displayed cone and its

boundary map. These identifications are identifications of perfect relative complexes: they can be computed with a locally free resolution of the one-arrow diagram and commute with derived base change.

At a geometric point, surface Serre duality gives

$$\mathrm{Ext}^2(I', Q)^\vee = \mathrm{Hom}(Q, I'(K)) = 0,$$

since Q is torsion and I' is torsion-free. The other possible Ext degrees lie in $[0, 1]$. The geometric-fiber criterion for a perfect complex proves the asserted Tor-amplitude. $\square$

5.2.2 Fixing the relative determinant

Let $\mathcal{P}_o^{\mathrm{der}}$ be the derived line stack rigidified along o , namely the fiber of evaluation along o over the trivial line. For any line M on a base-changed surface, its normalization is

$$M \otimes \pi_X^*(o^*M)^{-1}.$$

It has its canonical trivialization along o . Because $(\pi_X)_*\mathcal{O}_X = \mathcal{O}_S$ universally, all line automorphisms come from the base, and the normalization removes them. The classical truncation of $\mathcal{P}_o^{\mathrm{der}}$ is therefore the relative Picard scheme, parametrizing lines modulo base twists. Its tangent is

$$\mathbf{R}\pi_{X*}\mathcal{I}_o[1], \tag{5.6}$$

where constant functions split evaluation $\mathbf{R}\pi_{X*}\mathcal{O}_X \rightarrow \mathcal{O}_S$. This complex has geometric-fiber cohomology $H^1(\mathcal{O}_X)$ in degree zero and $H^2(\mathcal{O}_X)$ in degree one.

The determinant of perfect complexes is a functorial derived morphism [STV, Section 3.1]. Its differential is trace [STV, Proposition 3.2]. After normalization it maps the open arrow enhancement to $\mathcal{P}_o^{\mathrm{der}}$. Take its derived fiber at the normalized line $L_1 + L_2 - D$. Its truncation is exactly $\mathcal{Q}$: it fixes the Picard class of $\det I'$, not a trivialization of the unnormalized determinant on a parameter scheme. The relative tangent restricted to $\mathcal{Q}$ is

$$T_{\mathcal{Q}/S}^{\mathrm{vir}} = \mathrm{fib} \left\{ \mathbf{R}\mathrm{Hom}(I', Q) \xrightarrow{\partial, \mathrm{tr}, \mathrm{based}} \mathbf{R}\pi_{X*}\mathcal{I}_o[1] \right\}. \tag{5.7}$$

Here the trace identification is an identity of relative complexes, not an inference from equality at closed points. Locally represent a perfect family by a bounded complex $P^\bullet$ of finite free modules. Its determinant is $\bigotimes_j (\det P^j)^{(-1)^j}$. Differentiation on infinitesimal automorphisms gives the supertrace $\sum_j (-1)^j \mathrm{tr}_{P^j}$. The same calculation with square-zero complex coefficients gives the chain map $\mathbf{R}\mathrm{Hom}(P^\bullet, P^\bullet) \rightarrow \mathcal{O}$; it respects homotopies and change of perfect resolution. Using the identification of object tangents with shifted loop tangents gives its shift by $[1]$ as the determinant differential. This construction glues and commutes with base change, and therefore gives the relative trace in (5.7), including over nonreduced parameter schemes.

Lemma 5.3 (The determinant obstruction map). *Let $0 \rightarrow F \rightarrow V_0 \rightarrow Q_0 \rightarrow 0$ be an exact sequence on a smooth projective surface, where V_0 is a vector bundle of rank r and Q_0 is torsion. If K is ample and $Kc_1(Q_0) > rK^2$, the map*

$$\mathrm{Ext}^1(F, Q_0) \longrightarrow \mathrm{Ext}^2(F, F) \xrightarrow{\mathrm{tr}} H^2(\mathcal{O}_X)$$

is surjective.

Proof. By duality, a canonical form ω annihilating the image has the property that $\omega \text{id}_F : F \rightarrow F(K)$ extends to a map $a : V_0 \rightarrow F(K)$. If $i : F \rightarrow V_0$, the maps $i(K)a$ and ωid_{V_0} agree where $F = V_0$, hence everywhere. It follows that $\omega Q_0 = 0$. For nonzero ω , write $(\omega) = \sum a_j C_j$. At the generic point of C_j a quotient of $(\mathcal{O}/(\omega))^r$ has length at most ra_j . These generic lengths are the multiplicities of $c_1(Q_0)$. Thus $c_1(Q_0) \leq r(\omega)$ as divisorial cycles, and ampleness gives $Kc_1(Q_0) \leq rK^2$, a contradiction. The argument permits embedded points and nonreduced curve support. $\square$

Applying Lemma 5.3 with $r = 2$ and $D = 3K$ shows that the only possible degree-two cohomology of (5.7) vanishes. The complex is perfect of Tor-amplitude $[0, 1]$. The canonical map from the cotangent of a derived enhancement to that of its classical truncation has cone in degrees at most -2 [STV, Proposition 1.2 and Corollary 1.4]. Consequently the dual of (5.7) is a relative perfect obstruction theory in the sense of [BF, Definitions 4.4 and 5.1]. Adding the tangent of the smooth base by transitivity gives the absolute theory. All constructions are torus equivariant.

5.2.3 Virtual rank and compatible pullbacks

Riemann–Roch gives the relative virtual rank

$$v = \chi(I', Q) + \chi - 1 = 2\nu + (A_0 - K)D + \chi - 1 = 2t_0 + \chi - 3 > 2t_0. \quad (5.8)$$

Indeed, if $q_2 = \int \text{ch}_2 Q$, then $\chi(V, Q) = 2q_2 - (K + L_1 + L_2)D$ and $\chi(Q, Q) = -D^2$, while the rank of (5.6) is $1 - \chi$. Thus fixing determinant contributes the extra $\chi - 1$ in (5.8). The inequality $v > 2t_0$ will force the marked localization relation.

To compare the fixed virtual class with simpler classes, we need morphisms of obstruction theories, rather than only tangent classes in K-theory. The derived enhancements supply these morphisms as follows. For a morphism of these derived enhancements, transitivity maps to the transitivity triangle on truncations. If the two absolute comparison cones are C_Y, C_Z , the relative comparison cone fits into

$$Lf^*C_Z \longrightarrow C_Y \longrightarrow C_{Y/Z} \longrightarrow Lf^*C_Z[1].$$

The absolute cones lie in degrees at most -2 ; derived pullback preserves this upper bound and $[1]$ lowers it. Therefore $C_{Y/Z}$ also lies in degrees at most -2 . Whenever the relative derived cotangent has perfect amplitude $[-1, 0]$, this diagram supplies the compatible triple of obstruction theories. Virtual-pullback functoriality then applies [Man, Definition 4.5, Theorem 4.8, and Corollary 4.9]; see also [STV, Proposition 1.5].

5.3 Fixed loci and their virtual classes

The equivariant embedding and resolution hypotheses of virtual localization [GP, Section 1] hold here. The Grassmannian construction of relative Quot is equivariant and embeds the determinant fiber in a smooth scheme over the smooth base S . One can construct a global equivariant two-term presentation of its perfect obstruction theory using a linearized ample line. For completeness, over an affine base sufficiently positive twists generate its degree-zero cohomology and kill the higher cohomology of its degree-minus-one cohomology. Lifting generators gives a bundle mapping to the complex and surjective on degree-zero cohomology; its derived fiber is a bundle by the Tor-amplitude criterion. Further Serre vanishing lifts the map to the two-term embedding cotangent presentation. Torus linear reductivity makes the choices equivariant. For a nonaffine S , the same argument may be made after a Jouanolou affine-bundle pullback [ADO, Proposition 4.1.4(1)]. The resulting identity

descends by equivariant Chow homotopy invariance and smooth pullback of the base-changed obstruction theories. Subsequent dimension arguments are on the original base S .

A torus-fixed subsheaf of $L_1 \oplus L_2$ splits into its two weight subsheaves, also in families. Kernels and quotients in the two weights are flat, since they are direct summands of flat families. Thus a fixed point is a pair of rank-one arrows, subject to the total determinant constraint. Their direct-sum morphism of derived enhancements identifies its tangent triangle with the weight-zero part of (5.7), compatibly with the cotangent comparison. Normalizing a determinant cancels all character lines pulled from the base, so the based condition does not acquire a moving term. The moving virtual normal complex is

$$\bigoplus_{i \neq j} \mathbf{R} \operatorname{Hom}(I'_i, Q_j) \quad (5.9)$$

with weight $w_j - w_i$.

We use the divisor-and-residual-point factorization established in the relative setting by [Fog, Theorem 1.4]. We spell out its compatibility with the determinant and nonreduced bases. Each rank-one kernel has a unique factorization

$$I'_i = L_i(-\mathcal{C}_i)\mathcal{I}_{Z_i}, \quad D_i = [\mathcal{C}_i] \text{ in } \operatorname{Pic}(X/S), \quad D_1 + D_2 = D. \quad (5.10)$$

Here $\mathcal{C}_i$ is a relative effective Cartier divisor and Z_i is a finite-flat point family. This holds over nonreduced parameter schemes. To see this, let T be such a parameter scheme and untwist the target line. The T -flat kernel ideal I has a length-one locally free resolution, by the fiberwise projective-dimension bound. Its determinant is a line bundle on X_T , commuting with arbitrary base change. The maximal-minor map $I \rightarrow \det I$ restricts on every surface fiber to the inclusion of a rank-one torsion-free sheaf into its double dual. It is therefore fiberwise injective with zero-dimensional cokernel. Since both its source and target are T -flat, the local flatness criterion makes the map injective and its cokernel $\mathcal{T}$ T -flat. Properness and the fiber support make $\mathcal{T}$ finite over T .

The inclusion into the original target line extends uniquely across that finite support: on a smooth relative surface a finite-flat module has grade at least two, as follows by lifting a fiber regular sequence with flat successive quotients. Thus its Hom and first Ext into the structure sheaf vanish. Applying Hom to $0 \rightarrow I \rightarrow \det I \rightarrow \mathcal{T} \rightarrow 0$ gives the extension. The extension is fiberwise an injection of lines with Cartier quotient; the local flatness criterion makes its divisor relative Cartier. It gives the required point ideal and divisor, uniquely and compatibly with base change.

Let the point lengths be r, s . The Chern character of the two quotients gives

$$r + s = \nu + (A_0 - D_1)D_2. \quad (5.11)$$

The fixed scheme has only finitely many numerical choices. Its summand Hilbert polynomials are locally constant. In addition, the determinant line of each perfect kernel is a base-change-compatible line family, so its fiber Hilbert polynomial is locally constant. The residual point length is the difference between the Euler characteristics of this determinant and its kernel, and is therefore locally constant as well. Finite type now gives finiteness of these choices. All divisor and Picard moduli used in a choice may be restricted to the required finite-type quasi-projective pieces. For instance, lines of a fixed Hilbert polynomial form an open in the moduli of stable rank-one sheaves; rigidification at o represents their Picard classes.

Choose a *high* index with $KD_i > K^2$, which exists since $KD_1 + KD_2 = 3K^2$, and call the other index *low*. Write $P = D_{\text{low}}$. For this numerical choice, denote the fixed locus by $\mathcal{F}$ and the unrestricted low rank-one Quot scheme by $\mathcal{Q}_{\text{low}}$. Forgetting the high arrow gives a map

$$f : \mathcal{F} \longrightarrow \mathcal{Q}_{\text{low}}.$$

Its relative derived tangent is

$$\mathrm{fib} \left\{ \mathbf{R} \mathrm{Hom}(I'_{\mathrm{high}}, Q_{\mathrm{high}}) \longrightarrow \mathbf{R} \pi_{X*} \mathcal{I}_o[1] \right\}.$$

Lemma 5.3, now with rank one, gives amplitude $[0, 1]$. The transitivity comparison above is therefore a compatible triple. Consequently

$$[\mathcal{F}]^{\mathrm{vir}} = f^! [\mathcal{Q}_{\mathrm{low}}]^{\mathrm{vir}}. \quad (5.12)$$

Here $f^!$ is a bivariant virtual pullback. We will use its action on cycle representatives of the low virtual class, including cycles supported on special divisor loci.

Obstruction Euler factors of point Hilbert schemes also enter the surface Quot calculations of Oprea and Pandharipande [OP, Lemma 34 and Section 5.2]. We establish the relative comparison with varying determinant and arbitrary irregularity needed here.

Lemma 5.4 (Point excess). *The unrestricted low rank-one Quot scheme is classically*

$$\mathrm{Div}_P(X/S) \times_S X_{/S}^{[r_{\mathrm{low}}]},$$

where the divisor scheme permits its line to vary in the prescribed Picard data. Its virtual class is the divisor virtual class times the smooth relative point-Hilbert class, capped with

$$e(F_{\mathrm{low}}), \quad F_i = (\pi_* \mathcal{O}_{Z_i}(K - \mathcal{C}_i))^{\vee}. \quad (5.13)$$

Proof. The divisor enhancement parametrizes invertible-source arrows. Its tangent is $\mathbf{R} \pi_* \mathcal{O}_{\mathcal{C}}(\mathcal{C})$, of amplitude $[0, 1]$. Tensoring its universal source with the universal point ideal gives a morphism from this enhancement times the smooth relative Hilbert scheme to the rank-one arrow enhancement, with identity truncation. The smoothness of the relative point Hilbert scheme follows from the smooth surface Hilbert theorem [Fog, Theorem 2.4]. Indeed, choose disjoint etale surface charts around the finite support, with etale maps to the relative affine plane. Finite-subscheme deformations lift uniquely through these etale maps. Thus locally on the Hilbert functor the question is the product of the corresponding plane Hilbert functors over S , which is smooth over S .

Compare first the source objects of the arrows. The scalar trace summand of $\mathbf{R} \mathrm{Hom}(\mathcal{I}_Z, \mathcal{I}_Z)$ splits off. Writing $A = \mathbf{R} \pi_* \mathcal{O}_X$ and denoting the trace-free summand by B_Z , the source-object tangent is therefore $A[1] \oplus B_Z[1]$. The following calculation identifies $B_Z[1]$ with the relative Hilbert tangent, while retaining the entire Picard summand $A[1]$. The trace-free Ext groups in degrees zero and two vanish: an endomorphism extends to the determinant line, and surface duality gives the degree-two assertion. Riemann–Roch leaves a trace-free degree-one space of dimension $2r_{\mathrm{low}}$. The point-ideal boundary injects $\mathrm{Hom}(\mathcal{I}_Z, \mathcal{O}_Z)$ into this space, since every map $\mathcal{I}_Z \rightarrow \mathcal{O}_X$ is a scalar inclusion with zero composite into $\mathcal{O}_Z$. This boundary is the Hilbert Kodaira–Spencer map; its trace is zero because the point ideal has trivial determinant. Smoothness and the dimension $2r_{\mathrm{low}}$ make it an isomorphism. The family complexes are perfect, so the fiberwise isomorphism is an isomorphism of complexes after including the scalar line-moduli tangent.

For the arrow directions, the comparison is

$$\mathbf{R} \mathrm{Hom}(\mathcal{O}_X, \mathcal{O}_X(\mathcal{C})) \longrightarrow \mathbf{R} \mathrm{Hom}(\mathcal{I}_Z, \mathcal{O}_X(\mathcal{C})).$$

Its cone is

$$\mathbf{R} \mathrm{Hom}(\mathcal{O}_Z, \mathcal{O}_X(\mathcal{C}))[1] \simeq F_{\mathrm{low}}[-1]$$

by relative Serre duality. The latter is a genuine vector bundle in degree one: Z is finite flat and its direct image has no higher cohomology. We have therefore obtained a morphism of tangent theories whose cone is this excess obstruction bundle.

Because it comes from the morphism of enhancements, the dual comparison commutes with both obstruction morphisms to the same classical cotangent complex. The smaller obstruction bundle stack embeds in the larger with quotient F_{low} , and the intrinsic-cone embedding factors through the smaller stack. Zero-section excess intersection therefore multiplies the smaller virtual class by $e(F_{\text{low}})$. This proves the assertion, including when the divisor scheme is singular. $\square$

Let

$$\rho : \mathcal{Q}_{\text{low}} = \text{Div}_P(X/S) \times_S X/S^{[r_{\text{low}}]} \longrightarrow \text{Div}_P(X/S)$$

be the smooth projection. Combining point excess with (5.12) gives the fixed-cycle formula

$$[\mathcal{F}]^{\text{vir}} = f^! \left(e(F_{\text{low}}) \cap \rho^* [\text{Div}_P(X/S)]^{\text{vir}} \right). \quad (5.14)$$

This formula separates the two remaining tasks. We first determine where the low divisor virtual cycle can be supported. Over the surviving support, the high map f becomes smooth, and its virtual pullback will supply the second point Euler factor.

5.4 Support of the low divisor class

Put $y = KP/s_K$. The virtual relative dimension of the divisor scheme is

$$\frac{P(P-K)}{2}. \quad (5.15)$$

If $0 < y < 1$, this is at most $-(b-1)$. To check the integer bound, put $q = b-1$. The positive intersection $p = KP$ is a multiple of $2q$: each of the two canonical divisors on the product has this fiber degree. Thus

$$2q \leq p \leq 8q^2 - 2q, \quad \frac{P(P-K)}{2} \leq \frac{p^2/(8q^2) - p}{2} \leq -q + \frac{1}{4}$$

by Hodge index and convexity in p . The left side is integral, proving the claim. The absolute divisor virtual cycle consequently has a representative of dimension at most $\dim S - (b-1)$. The image of its support in S has codimension at least $b-1$. In (5.14), smooth pullback by ρ and cap product with $e(F_{\text{low}})$ preserve this base support. So does $f^!$: it is bivariant and commutes with restriction and proper pushforward [Man, Theorems 4.1 and 4.4]. Although these pullbacks can increase the dimension of a cycle, they do not enlarge the locus over which it lies in S . Every marked fixed contribution above this low cycle therefore vanishes in our support quotient.

On the Picard open where $H^2(P) = 0$, consider instead the morphism from the divisor enhancement to the derived based Picard scheme. The divisor exact sequence shows that its obstruction map surjects onto $H^2(\mathcal{O}_X)$. Hence its relative tangent has amplitude $[0, 1]$, and the transitivity construction again gives a compatible virtual-pullback triple. The absolute virtual dimension of that Picard theory is

$$\dim S + 1 - \chi < 0.$$

Its virtual class is zero, so the restricted divisor virtual class is zero as well. For $y > 1$, the condition $H^2(P) = 0$ is automatic: otherwise duality makes $K - P$ effective, contrary to ampleness. For $y = 1$, a nonzero section of $K - P$ has zero intersection with K , and therefore has empty divisor. Thus the exceptional line is exactly K . Chow localization expresses the divisor virtual class as a pushforward from the canonical linear system $|K|$. For $y = 0$, an effective divisor is empty; its tangent complex is zero and its class is the base fundamental class.

We have reduced the low divisor support to 0 or K , and hence the high line to $3K$ or $2K$. On a neighborhood of either support the high direction is smooth: Kodaira vanishing gives its projective bundle of sections, together with its smooth point Hilbert scheme. The relative virtual tangent class there is

$$[\mathbf{R}\pi_*\mathcal{O}(\mathcal{C}_{\text{high}})] - 1 - [F_{\text{high}}] + [T_{X^{[r_{\text{high}}]}/S}]. \quad (5.16)$$

Indeed, the unrestricted rank-one arrow class subtracts $\mathbf{R}\pi_*\mathcal{O}_X$, while fixing the based determinant subtracts $\mathbf{R}\pi_*\mathcal{I}_o[1]$; together these subtract exactly one. The remaining point-ideal calculation is that in Lemma 5.4. The projective Euler sequence identifies actual tangent minus (5.16) with $[F_{\text{high}}]$.

For a smooth morphism with a relative perfect obstruction theory, the obstruction sheaf is locally free and its virtual pullback is smooth pullback capped with its top Chern class. This follows by locally splitting the cotangent comparison onto the locally free relative cotangent; equivalently, the intrinsic cone is $[0/T_{\text{rel}}]$ and zero-section excess gives the Euler factor. The argument works on arbitrary base cycles, not just the fundamental cycle; compare [BF, Proposition 7.3] and [Man, Theorem 4.4]. Here the obstruction bundle has K-class $[F_{\text{high}}]$, so its Euler class is $e(F_{\text{high}})$. Let $\mathcal{D} \subset \text{Div}_P(X/S)$ be the empty-divisor locus S or the canonical linear system $|K|$. Base-change f to $\mathcal{D} \times_S X_{/S}^{[r_{\text{low}}]}$, and use the same letters f, ρ for the resulting maps. For every cycle α on this product, the high virtual pullback is therefore

$$f^!(\alpha) = e(F_{\text{high}}) \cap f^*(\alpha). \quad (5.17)$$

In particular, apply it to $\alpha = e(F_{\text{low}}) \cap \rho^*\beta$, where β is a cycle on $\mathcal{D}$ supplied by Chow localization for the low divisor virtual class. Bivariant proper-pushforward compatibility then returns the result to $\mathcal{F}$. All operational insertions restrict to the actual linear systems on those cycles; no formula for the low cycle supported on $|K|$ is needed. These linear systems are smooth over S : for K , constant cohomology dimensions follow from the product of curves and base change, and for $2K, 3K$ from Kodaira vanishing. Their universal divisor classes are $D_i + x_i$, where x_i is the hyperplane class.

5.5 Marked localization and the short-length exclusion

Adjoin $m = 3t_0$ independent surface positions to $\mathcal{Q}$. At each position insert $\text{ch}_2^{\mathbb{G}_m}(Q)$, cap with the smoothly pulled virtual class, and push to $X_{/S}^m$. This push is proper. The torus acts trivially on the output. Before localization the total codimension is $2m - v$. An ordinary output component of codimension $4t_0$ would therefore have equivariant exponent

$$(2m - v) - 4t_0 = 2t_0 - v < 0.$$

It is absent. Thus the ordinary codimension- $4t_0$ part of the localized fixed sum is zero after setting the equivariant parameter equal to one.

On a fixed branch the insertion is a sum, for $i = 1, 2$, of

$$[Z_i] + (L_i + w_i)(D_i + x_i) - \frac{(D_i + x_i)^2}{2}. \quad (5.18)$$

Here $[Z_i]$ denotes the incidence cycle on the corresponding surface position. Test every surface slot by the even tensor

$$z_\gamma = \sum_{\alpha, \beta} \gamma_{\alpha\beta} w_\alpha \otimes w_\beta. \quad (5.19)$$

This is a tensor test in the coefficient algebra, not a numerical intersection test alone. Since a unit curve class pairs to zero with a pure W_b input, a line-only insertion that survives this test uses a

positive divisor degree on both curve axes. Hence each such position supplies at least two ordinary codimensions, from powers of the two curve canonical classes.

Suppose k of the positions use line-only insertions. Extract these $2k$ divisor codimensions by the projection formula. The remaining $m - k$ incidence positions lie in the union of the two point subschemes, of total length $r + s$. Their pushed support therefore has at most $r + s$ distinct surface positions and has codimension at least

$$2(m - k - r - s)$$

in those factors. Operational incidence classes vanish away from their incidence supports, so this support assertion applies to successive caps with arbitrary low cycle multipliers. If $r + s < t_0$, the displayed codimension is strictly larger than the available $4t_0 - 2k$, and the tested contribution is zero.

By (5.11), the four remaining ordered divisor pairs have lengths

$$\begin{array}{c|cccc} (D_1, D_2) & (0, 3K) & (3K, 0) & (K, 2K) & (2K, K) \\ \hline r + s & \nu + 3s_K & \nu & \nu & \nu - s_K. \end{array}$$

Only $(D_1, D_2) = (0, D)$ survives. On this branch the low divisor is empty and its virtual class is the base class. The calculation is on the product of the two point Hilbert schemes and $|D|$, with both point Euler factors and the ordinary fundamental cycle. We have

$$N = r + s = \nu + ds_K, \quad \dim(|D|/S) = \chi + \frac{d(d-1)}{2}s_K - 1 = t_0, \quad x_1 = 0, \quad x_2 = x. \quad (5.20)$$

5.6 A relative marked Hilbert-scheme recursion

The surviving fixed contribution must be expanded as a Chow class on $X_{/S}^m$, with every output retained. Relative EGL recursion for surface pushforwards to a base appears in [Kong, Section 2.1, Lemma 2.1 and Proposition 2.2]; the same work also treats one and two marked curve outputs [Kong, Section 3.3, Lemma 3.4]. The construction below retains an arbitrary finite set of surface outputs. This is the form that lets the codimension bound identify the Chow tensor left by localization.

Lemma 5.5 (Marked universality). *Fix finitely many marked output surface factors. Push forward, over relative point Hilbert schemes and any additional surface factors, a finite polynomial in Chern characters of universal point ideals, their derived duals, line classes, relative tangent Chern classes, and equality diagonals. The resulting Chow class is a universal finite sum of decorated equality diagonals among the outputs, multiplied by relative characteristic pushes from closed surface components. One fixed polynomial procedure chooses the expression, independently of the surface family. No marked output is integrated in the procedure.*

Proof. For a positive point length r , use the consecutive nested scheme

$$\mathcal{N}_r = X_{/S}^{[r-1, r]} \xrightarrow{\psi} X_{/S}^{[r]}, \quad \sigma = (\phi, \rho) : \mathcal{N}_r \longrightarrow X_{/S}^{[r-1]} \times_S X.$$

It is the quotient projectivization of the smaller universal ideal $\mathcal{I}_{r-1}$. The universal quotient line $\mathcal{L}$ and the residual graph j give

$$0 \longrightarrow \psi_X^* \mathcal{I}_r \longrightarrow \phi_X^* \mathcal{I}_{r-1} \longrightarrow j_* \mathcal{L} \longrightarrow 0. \quad (5.21)$$

The consecutive nested scheme and point Hilbert schemes are smooth over S . These are the smooth-surface facts underlying [EGL, Section 1]. For the relative assertion, choose disjoint etale

charts about the finite support. Deformations of a finite subscheme, and of an inclusion of two such subschemes, lift uniquely under an étale map to the relative plane. The resulting local deformation functors are products of the plane functors, base-changed from the ground field to S ; their smoothness gives the assertion. Over the distinct-point locus ψ has degree r , so $\psi_*[\mathcal{N}_r] = r[X/S^{[r]}]$. Pullback followed by pushforward and division by r therefore preserves the original calculation with all other factors present.

Put $\zeta = c_1(\mathcal{L})$. In every surface slot, replace the ideal by (5.21) in K-theory. GRR for the residual graph gives

$$\mathrm{ch}(j_*\mathcal{L}) = j_*(e^\zeta \rho^* \mathrm{td}(T_{X/S})^{-1}).$$

The graph is a regular codimension-two embedding with the indicated normal bundle. Thus its only new ingredients are an equality diagonal to the residual surface coordinate, ζ , and relative tangent classes. Derived duals give no new ingredients, since $\mathrm{ch}_j(E^\vee) = (-1)^j \mathrm{ch}_j(E)$.

Choose a two-term locally free resolution

$$0 \longrightarrow V_- \longrightarrow V_+ \longrightarrow \mathcal{I}_{r-1} \longrightarrow 0.$$

The resolution exists by the fiberwise projective-dimension bound and an ample locally free presentation. The quotient projectivization is the zero locus of $V_-^\vee(1)$ on $\mathbb{P}(V_+)$. Its codimension is $\mathrm{rk} V_-$, by the dimension of the consecutive nested scheme, so the section is regular. The projective bundle formula then gives

$$\sigma_*(\zeta^k \cap [\mathcal{N}_r]) = \left[\frac{c(V_-^\vee)}{c(V_+^\vee)} \right]_k \cap [X/S^{[r-1]} \times_S X]. \quad (5.22)$$

This is the resolution Segre rule of [EGL, Section 1, Lemma 1.1]. After substituting the universal sequence, the integrand is a polynomial in ζ with coefficients pulled from the smaller Hilbert scheme, the residual point, and the unchanged factors. Formula (5.22) lowers the total Hilbert length. Repeated application ends at length zero.

Products of surface pushes are first rewritten using separate surface slots, by flat base change and the projection formula. The same applies to GRR expressions for point-ideal Hom complexes and tautological bundles, so it covers all the Euler factors that will occur here after the finite Chow-degree truncation. At length zero only equality diagonals and the specified decorations remain. An equality edge connecting two previously distinct components identifies their coordinates transversely. An edge within an already connected component contributes the excess class $c_2(T_{X/S})$. Each connected component meeting outputs gives their equality diagonal with all its decorations on the common coordinate; a component meeting no output is integrated relatively over X . Fixing the order of reductions and equality edges gives one universal polynomial procedure throughout. $\square$

Apply Lemma 5.5 to the surviving fixed branch. For the surface product of curves, all the allowed output decorations expand into powers of the two curve canonical classes. Consider a connected equality block containing k outputs. For $k = 1$, pure-slot testing requires at least two decorating codimensions, one on each axis. For $k = 2$, a bare diagonal on either curve axis pairs two isotropic inputs and is zero. Decorations are therefore needed on both axes, in addition to the codimension-two surface diagonal. For $k \geq 3$, the equality block already costs $2(k-1)$ codimensions. These lower bounds are summarized in Table 2. Every surviving block costs at least $4/3$ codimensions per output, with equality only for an undecorated three-output block.

The total selected codimension is $4t_0$ on $3t_0$ outputs. It follows that only undecorated triple blocks occur. No line-only insertion survives. There is no remaining positive horizontal codimension: every closed surface component must have top relative degree, and every base bundle contributes

Outputs	Requirement under isotropic testing	Codimension at least
1	A divisor on each curve axis	2
2	Equality diagonal and a divisor on each axis	4
$k \geq 3$	Surface equality diagonal	$2(k-1)$

Table 2: The codimension budget with all output positions retained. Only a bare triple attains cost $4/3$ per output.

only its scalar characteristic term. Integration over $|D|$ therefore extracts exactly x^{t_0} . In particular, the scalar multiplying the triple blocks can be computed over a point with the same numerical characteristic data, using the fixed polynomial procedure of Lemma 5.5.

5.7 The scalar interface

To specify this scalar without an independence assumption on diagonal classes, use the polynomial procedure just fixed. For a formal surface input z , write a connected component carrying k marked inputs and decoration α as the moment

$$M_k(\alpha) = \int_X z^k \alpha.$$

Give unmarked moments their actual top intersection numbers. Give marked moments value zero except for $M_3(1) = 1$. Thus this evaluation retains precisely bare triple blocks. In the generating series of Section 6, the variable q records total point length, p is the exponential variable for marked positions, and x is the high hyperplane class. Let a_{N,t_0} be the coefficient defined in (6.4), with the moments evaluated in this way and N as in (5.20). Equivalently, $(3t_0)! a_{N,t_0}$ is the sum of the coefficients of all bare triple partitions in the labeled marked pushforward. The polynomial procedure is fixed before imposing relations among the diagonal classes of a particular surface. Section 6 evaluates this same representative: its moment-additivity argument is what permits numerical surface calculations to determine the selected scalar. Thus the passage to those calculations does not replace the marked Chow identity by numerical universality.

Proposition 5.6 (Localization relation). *Under the hypotheses of Theorem 5.1, the scalar a_{N,t_0} satisfies*

$$a_{N,t_0} T_\gamma^{t_0} = 0 \quad \text{in } \mathfrak{B}_{S,b-1}.$$

Proof. The codimension- $4t_0$ marked localization sum is zero. The low-divisor and short-length arguments discard every fixed contribution except the branch $(0, D)$, in the stated support quotient. Marked universality and the output-degree bound leave only its bare triple partitions, with total coefficient $(3t_0)! a_{N,t_0}$.

Test each surface position with z_γ from (5.19). A bare surface triple diagonal is the product of the two curve triple diagonals, hence evaluates to T_γ up to the Koszul sign from grouping the two curve axes. This sign is the same for every block and every triple partition: each whole surface input is even, so permuting those inputs introduces no further sign. Thus every partition evaluates to the same signed $T_\gamma^{t_0}$. Dividing the resulting equality by its common sign and by $(3t_0)!$ proves the assertion. All equalities were Chow equalities or tensor tests in the faithful coefficient formalism of Proposition 4.1. $\square$

6 The triple coefficient and arbitrary color products

We first compute the scalar left by Proposition 5.6, completing the surface bound. We then convert that bound on isotropic tests into vanishing for sufficiently long products with arbitrary color inputs. For the coefficient calculation, put

$$t = t_0, \quad N = t - 1 + ds_K, \quad A_0 = K, \quad D = dK, \quad d = 3, \quad B = -A_0 - D.$$

The letter q in the generating series below is a formal variable. It is unrelated to the realization maps $q_{g,n}$.

Proposition 6.1. *The bare-triple scalar a_{N,t_0} in Proposition 5.6 is nonzero. Consequently Theorem 5.1 holds.*

The calculation belongs to the study of tautological Hilbert-scheme integrals developed by Lehn and Ellingsrud–Göttsche–Lehn [Leh, Section 4.3], [EGL]. For the symmetric-power and residue methods used below, compare [MOP2, Equation (9), Theorem 2, and Lemma 3]. The coupling of two Hilbert factors, all marked moments, and the Euler-shift derivative will be retained explicitly.

We first identify a generating series for this scalar. We then compute its first nonzero hyperplane coefficient and prove that the resulting one-variable coefficient cannot vanish.

6.1 The moving factor

For a vector bundle E , write $e_a(E)$ for its Euler class after adding a to each Chern root. If a has invertible constant term, this convention extends multiplicatively to virtual bundles and perfect complexes. Inverses are expanded in positive Chow degree. Let

$$M_{r,s} = X^{[r]} \times X^{[s]},$$

with universal finite subschemes Z_1, Z_2 on $M_{r,s} \times X$. Derived Hom expressions in this subsection include pushforward along the surface X .

On the surviving fixed component, the two kernels are

$$I'_1 = L_1 \mathcal{I}_{Z_1}, \quad I'_2 = L_2(-D - x) \mathcal{I}_{Z_2},$$

where x is the hyperplane class of $|D|$. Apart from the factor $(1 - x)^{\chi(B)}$, the inverse moving Euler class is

$$\begin{aligned} R(x) = & \frac{e_1(((K + A_0)^{[r]})^\vee)}{e_{1-x}(((K - B)^{[r]})^\vee) e_{1-x}(B^{[s]}) e_{-1+x}((-B)^{[r]})} \\ & \times e_{1-x}(R\mathrm{Hom}(\mathcal{O}_{Z_1}, \mathcal{O}_{Z_2}(B))) e_{1-x}(R\mathrm{Hom}(\mathcal{O}_{Z_1}, \mathcal{O}_{Z_2}(K + B))). \end{aligned} \quad (6.1)$$

Here and below only scalar terms of bundles pulled from the original base can contribute, by Proposition 5.6. In particular the omitted inverse Euler factor of $R\pi_*(-A_0)$, at weight one, has scalar term one.

To check (6.1), the forward moving arrow has virtual tangent

$$R\mathrm{Hom}(\mathcal{I}_{Z_1}, -A_0) - R\mathrm{Hom}(\mathcal{I}_{Z_1}, \mathcal{I}_{Z_2}(B)),$$

at weights 1 and $1 - x$, respectively. Expand each point ideal as $\mathcal{O} - \mathcal{O}_Z$. The constant term of the second Hom gives $(1 - x)^{\chi(B)}$. Surface duality identifies

$$R\mathrm{Hom}(\mathcal{O}_{Z_1}, B) \quad \text{with} \quad ((K - B)^{[r]})^\vee[-2]$$

in the convention relevant to the Euler class, and likewise for $-A_0$. This gives the first numerator and the first two denominator factors of (6.1). The reverse moving arrow is

$$R\mathrm{Hom}(\mathcal{I}_{Z_2}, \mathcal{O}_{Z_1}(-B))$$

at weight $-1 + x$. Its point term gives the final denominator. Duality changes its cross term to the remaining numerator term. The cross Hom has virtual rank zero, so changing both its dual and the sign of its weight introduces no rank sign.

The fixed obstruction Euler classes on this component are

$$e((K^{[r]})^\vee), \quad e_x(((K - D)^{[s]})^\vee).$$

The sign of the second shift is worth recording: the line in the obstruction bundle is $K - D - x$ before taking the dual, and hence has twist $+x$ afterwards.

6.2 Marked moments and shifted additivity

Temporarily replace K and $K - D$ in these two fixed Euler classes by independent divisor variables L, H . Also separate the Euler shift x_e from the moving shift x_m . For a surface class z , let

$$\mu_r(z) = (\mathrm{pr}_{X^{[r]}})_*([Z_1] \mathrm{pr}_X^* z),$$

and define $\mu_s(z)$ similarly. Consider

$$\Theta(q, p, x_e, x_m) = \sum_{r,s \geq 0} q^{r+s} \int_{M_{r,s}} e((L^{[r]})^\vee) e_{x_e}((H^{[s]})^\vee) R(x_m) \exp(p(\mu_r(z) + \mu_s(z))). \quad (6.2)$$

The exponential records labeled output insertions with their usual factorial denominators.

We use the fixed marked recursion of Proposition 5.6 to regard each finite coefficient of (6.2) as a polynomial in decorated moments

$$M_k(\alpha) = \int_X z^k \alpha.$$

Here α is a monomial in $L, H, A_0, D, K, c_2(T_X)$. More precisely, carry out the nested-Hilbert recursion while retaining the marked output slots. Each connected component of the final equality graph is integrated over one copy of X ; if it contains k marked slots and decoration α , replace that integration by $M_k(\alpha)$. This fixes a polynomial representative before imposing numerical relations between moments. The recursion and its projective-bundle signs are those of [EGL, Sections 1 and 3]. The version with arbitrary retained outputs used here is proved in Lemma 5.5. This is the interface that keeps the calculation in relative Chow throughout.

Let $\mathcal{E}$ denote the following evaluation:

$$\mathcal{E}(M_0(\alpha)) = \int_X \alpha, \quad \mathcal{E}(M_3(1)) = 1, \quad \mathcal{E}(M_k(\alpha)) = 0 \quad (k > 0, (k, \alpha) \neq (3, 1)). \quad (6.3)$$

Unmarked integrations in this formula are their top-degree numerical integrals; other degrees give zero. The scalar normalization agreed with Proposition 5.6 is

$$a_{N,t_0} = [q^N p^{3t} x^t] (1 - x)^{\chi(B)} \mathcal{E}(\Theta(q, p, x, x))|_{L=K, H=K-D}. \quad (6.4)$$

Indeed, $\mathcal{E}$ retains precisely the partitions into bare triples. Passing from the exponential series to labeled outputs multiplies by the nonzero common factor $(3t)!$. Every such partition has the same tested value, up to the common Koszul sign for grouping the two curve axes. Those factors have already been divided out in Proposition 5.6.

The next observation allows us to compute (6.4) using ordinary numerical surface calculations, without assuming that all formal moments are independent.

Lemma 6.2. *There is a series $c(q, x_e, x_m)$ such that*

$$\mathcal{E}(\Theta) = \mathcal{E}(\Theta|_{p=0}) \exp(c(q, x_e, x_m)p^3/3!). \quad (6.5)$$

Moreover, let $c_H(q)$ be the coefficient of $M_3(H)$, with the factor $p^3/3!$ removed, in the one-moment part of $\log \Theta(q, p, 0, 0)$. Then

$$[x_e]c(q, x_e, 0) = -c_H(q). \quad (6.6)$$

Proof. Fix a finite coefficient cutoff, so that all polynomials and moment spaces in this argument are finite-dimensional. On genuine numerical data, allowing any even inhomogeneous cohomology class z , Θ is multiplicative under disjoint union of surfaces. The Hilbert schemes split according to the lengths on the components; all bundles and Hom complexes split; shifted Euler classes are multiplicative on direct sums; and the incidence exponentials multiply. These statements hold with both x_e and x_m retained.

Consequently the polynomial for $\log \Theta$ is additive on nonnegative integral combinations of genuine moment vectors. For any finite collection of those vectors, polynomial interpolation on the nonnegative integer grid extends additivity to their rational linear span. In particular, its homogeneous components of moment-count at least two vanish on that span. This is the disjoint-union argument underlying [EGL, Theorem 4.2], applied coefficientwise to the specified marked polynomial.

The vector in (6.3) belongs to this span. Start with the genuine unmarked vector obtained by setting $z = 0$. On a fixed connected surface compare $z = u + c_0[\text{pt}]$ with $z = u$, for rational u, c_0 . The difference changes only the undecorated marked moments, by

$$M_k(1) \mapsto ku^{k-1}c_0.$$

For every positive-degree decoration the change is zero. Varying u and using a finite Vandermonde matrix isolates $M_3(1)$ among all the moments within the cutoff. The logarithm therefore has, at this vector, only an unmarked term and a term proportional to p^3 , proving (6.5).

We next justify differentiating in the Euler shift. Use the universal polynomial before discarding decoration degrees larger than the dimension of the surface. In this representative, the substitution

$$H \mapsto H - x_e$$

computes the shifted Euler class, since

$$e_{x_e}((H^{[s]})^\vee) = (-1)^s e((H - x_e)^{[s]}).$$

This notation means a base-line twist, not subtraction of a numerical surface divisor. To realize it, use the constant surface family over $\mathbb{P}^a$, with a larger than the coefficient cutoff, and twist H by $\mathcal{O}(-x_e)$. GRR for the tautological bundle uses $\exp(H - x_e)$; the nested-Hilbert recursion and every projection formula commute with this twist and with moving powers of x_e out of the pushforward. It thus gives exactly the substitution in the unsimplified moment polynomial. The moving factor is independent of the auxiliary H , so its degree cutoff is unaffected. More explicitly, an output consisting of j bare triples requires total intrinsic Euler-and-ratio degree $2(r + s) - 2j$ on the Hilbert factors. The Euler factors have nonnegative intrinsic degrees. Thus moving terms of degree greater than $2(r + s)$ contribute neither to genuine integrals nor to the desired formal coefficient, before or after the twist.

It is important to apply additivity to the shifted invariant itself. The substitution is linear on the moment symbols and preserves their number in a monomial. For each x_e , the resulting

polynomial still computes the genuine shifted invariant and is additive after taking its logarithm. Hence every shifted component of moment-count at least two vanishes on the original genuine span, and in particular at $\mathcal{E}$. There is no claim that shifting the evaluation vector preserves that span. For example, an individual term involving $M_0(H^3)$ can acquire a nonzero derivative after shifting, but the nonlinear terms cancel collectively by this additivity.

In the one-moment part of p -degree three, the derivative followed by $\mathcal{E}$ can only come from

$$M_3(H - x_e) = M_3(H) - x_e M_3(1).$$

Other positive decorations remain positive after a single derivative, unless the decoration was H itself. Thus the derivative is $-c_H(q)$, proving (6.6). $\square$

6.3 Restriction to two curves

We first claim that

$$c(q, 0, x_m) = 0. \tag{6.7}$$

Represent very ample L, H by smooth transverse curves. At $x_e = 0$, the two Euler classes in (6.2) restrict the integral to $L^{(r)} \times H^{(s)}$, with sign $(-1)^{r+s}$. These are the regular zero loci of the tautological sections: the zero schemes are the Hilbert schemes of points on the two smooth curves, of the expected codimensions r, s . A point-class perturbation of the constant $z = u$ restricts to zero on both curves. The Vandermonde test in the preceding proof therefore annihilates the bare marked coefficient. The coefficient of the single moment $M_3(1)$ is a universal scalar, independent of the numerical line classes, so this proves (6.7) identically.

To determine c_H , set both shifts to zero and take

$$z = u + \tau Z$$

for a divisor Z on the surface. Write x', y' for the numerical point divisors on $L^{(r)}, H^{(s)}$. After the Euler restrictions, the one-body factors of (6.1), including their signs, are

$$c_{-1}((A_0 + 2D)^{[r]})^{-1} (-1)^s c_{+1}(B^{[s]})^{-1}. \tag{6.8}$$

Here $c_a(E) = \sum_j a^j c_j(E)$. For verification, on the ordered cover of a symmetric power, a tautological line bundle $M^{[r]}$ has a filtration with roots

$$M_i - \delta_i, \quad \delta_i = \sum_{j < i} [\Delta_{ij}].$$

Two line classes pulled from the same curve position multiply to zero. Applying this to the low ratio in (6.1) reduces its factor at position i to

$$\frac{1}{1 + \delta_i - (A_0 + 2D)_i},$$

which gives the first term in (6.8).

Put $I = LH$. At each of the I transverse intersections, the two universal finite subschemes meet by independently imposing their point-containment divisors. Their derived Hom has the class of the structure sheaf of the product of those divisors, twisted by the determinant normal of the first universal subscheme. Numerically its class is

$$e^{x'}(1 - e^{-x'})(1 - e^{-y'}).$$

Its Euler class at weight one is therefore

$$\frac{(1+x')(1-y')}{1+x'-y'}.$$

The two cross Hom terms give the combined factor

$$\left(\frac{(1+x')(1-y')}{1+x'-y'} \right)^{2I}. \quad (6.9)$$

This calculation also follows by resolving the two transverse containment divisors and taking their tensor-product resolution; the twists by surface lines have numerical degree zero at the fixed intersection point. The incidence exponential becomes

$$\exp(pu(r+s) + p\tau(ZLx' + ZHy')). \quad (6.10)$$

We recall the symmetric-power integral needed to sum these terms. For a curve of genus h , a line of degree d_1 , and its tautological bundle on the r -th symmetric power,

$$c_a(\text{line}^{[r]}) = (1-ax')^{r-d_1-1+h} \exp\left(\frac{a\vartheta}{1-ax'}\right), \quad \int (x')^{r-j} \frac{\vartheta^j}{j!} = \binom{h}{j}. \quad (6.11)$$

Here ϑ is the pulled-back Jacobian polarization. For completeness, the universal divisor sequence and GRR give

$$\text{ch}(\text{line}^{[r]}) = d_1 + 1 - h + (r - d_1 - 1 + h + \vartheta) \exp(-x').$$

Indeed the universal divisor class has terms $r[\text{pt}] + x' + \ell$, where the mixed class satisfies $\int_{\text{curve}} \ell^2/2 = -\vartheta$. Converting this Chern character to the total Chern class gives the first formula in (6.11). For the second, pull to the ordered cover and expand the polarization in a symplectic basis. Intersecting the point divisors first leaves j factors; choosing j distinct symplectic pairs contributes one after the factorials, giving $\binom{h}{j}$.

Let

$$d_1 = (A_0 + 2D)L, \quad d_2 = BH, \quad 2h_1 - 2 = L(L + K), \quad 2h_2 - 2 = H(H + K).$$

By (6.11), low-curve integration of the first factor in (6.8), times any series in x' , is coefficient extraction against

$$[(x')^r] (1+x')^{d_1+1-2h_1-r} (1+2x')^{h_1}. \quad (6.12)$$

High-curve integration has the analogous expression after replacing y' by $-y'$, which absorbs the sign $(-1)^s$. Accordingly (6.9) becomes

$$\left(\frac{(1+x')(1+y')}{1+x'+y'} \right)^{2I},$$

and the nonconstant part of (6.10) becomes $\exp(p\tau(ZLx' - ZHy'))$.

The elementary Lagrange identity used for each variable is

$$\sum_{r \geq 0} Q^r [x^r] \frac{F(x)}{(1+x)^r} = F(w) \frac{1+w}{1+2w}, \quad w(1+w) = Q, \quad w(0) = 0. \quad (6.13)$$

One proof is to express coefficient extraction as the formal residue at zero: summing the geometric series gives $F(x)(1+x)/(x(1+x) - Q)$, whose small root is w ; its residue there is the right side.

The identity holds with other formal variables in the coefficients, so it can be applied successively to x' and y' . Using $Q = q \exp(pu)$, both small roots equal w . Equations (6.12)–(6.13) therefore give

$$\begin{aligned} \log \Theta(q, p, 0, 0) &= (d_1 + d_2 + 4 - 2h_1 - 2h_2 + 4I) \log(1 + w) \\ &\quad + (h_1 + h_2 - 2 - 2I) \log(1 + 2w) + p\tau(ZL - ZH)w. \end{aligned} \quad (6.14)$$

This equation is the genuine numerical evaluation with $z = u + \tau Z$. It determines the universal coefficients used below.

6.4 The first hyperplane coefficient

Let $w_0(1 + w_0) = q$, with $w_0(0) = 0$, and let $\mathcal{D} = q\partial_q$. By Lemma 6.2, the numerical evaluation of $\log \Theta$ on a genuine moment vector equals that of its one-moment part: every component with at least two moment factors vanishes there. Its term of p -degree three is a universal scalar combination of the moments $M_3(\alpha)$. The term involving $M_3(H)$ is $c_H(q)M_3(H)p^3/3!$. Since

$$[u^2\tau]M_3(H) = 3ZH,$$

we can identify c_H by varying the very ample H , with the other divisor classes fixed. Only degree-one decorations can contribute to $u^2\tau$, and H is the only varying such decoration. For example, one may take arbitrarily large integral multiples of a fixed ample line for H , with Z fixed ample. The first two terms of (6.14) contain no τ . The coefficient of $p^3u^2\tau(ZH)$ in its last term is $-\mathcal{D}^2w_0/2$. Comparing factorials yields

$$c_H(q) = -\mathcal{D}^2w_0.$$

Together with (6.6) and (6.7), this proves

$$[x]c(q, x, x) = \mathcal{D}^2w_0 = \frac{q(1 + 2q)}{(1 + 4q)^{3/2}}. \quad (6.15)$$

There is no contribution from differentiating x_m , because $c(q, 0, x_m)$ is identically zero.

The unmarked calculation is equally explicit. Set $p = 0$ in (6.14). Every finite coefficient is polynomial in the intersection numbers of L, H, A_0, D, K , so the formula continues from very ample L, H to $L = K, H = K - D$. This can also be seen by varying sufficiently large multiples of K in both variables and then using polynomial identity. The adjunction expressions for h_1, h_2 are polynomial expressions in this continuation, not assertions that a negative line has a smooth effective representative. Substitution gives

$$\mathcal{E}(\Theta(q, 0, 0, 0))|_{L=K, H=K-D} = (1 + w_0)^{ds_K} (1 + 2w_0)^{d(d+1)s_K/2}. \quad (6.16)$$

6.5 A residue with a definite sign

We now prove Proposition 6.1. By (6.5), the p^{3t} -coefficient is $c(q, x, x)^t/(6^t t!)$ times the unmarked series. The series $c(q, x, x)$ has zero constant term in x . Thus the coefficient of x^t uses its linear term in every factor, and uses only the constant terms in x of the unmarked series and of $(1 - x)^{\chi(B)}$. Equations (6.4), (6.15), and (6.16) give

$$a_{N,t_0} = \frac{1}{6^t t!} [q^{ds_K-1}] (1 + w_0)^{ds_K} (1 + 2q)^t (1 + 4q)^E, \quad E = \frac{d(d+1)s_K}{4} - \frac{3t}{2}. \quad (6.17)$$

Put $v = 1 + 2w_0$, so that $q = (v^2 - 1)/4$. The coefficient residue in (6.17) becomes

$$2^{ds_K-t-1} \operatorname{Res}_{v=1} \frac{(1 + v^2)^t v^{2E+1}}{(v - 1)^{ds_K}} dv. \quad (6.18)$$

For the present parameters,

$$ds_K = 24\chi, \quad t = 25\chi - 1, \quad E = \frac{3 - 27\chi}{2}, \quad \chi = (b - 1)^2 \geq 16.$$

Hence the rational differential in (6.18) has finite poles only at zero and one. Its rational coefficient has degree $2 - \chi \leq -14$ at infinity, so its residue at infinity is zero. The residue at zero is

$$[v^{27\chi-5}](1 + v^2)^{25\chi-1}(1 - v)^{-24\chi}.$$

This is strictly positive: the second factor has positive coefficients in every nonnegative degree, the first has nonnegative coefficients and constant term one, and $27\chi - 5 \geq 0$. The residue at one is its negative and therefore is nonzero. Both scalar prefactors in (6.17)–(6.18) are nonzero. This proves $a_{N,t_0} \neq 0$.

Finally, Proposition 5.6 gives $a_{N,t_0}T_\gamma^{t_0} = 0$ for the isotropic cubic contraction T_γ appearing in Theorem 5.1. The scalar is a nonzero rational number and is invertible in the coefficient algebra. Thus $T_\gamma^{t_0} = 0$, which completes the proof of that theorem.

6.6 From isotropic tests to arbitrary color products

The surface theorem now gives nilpotence for every isotropic tensor test. To use it in the product defining the counterexample, we need a vanishing statement for arbitrary color inputs. We return to a smooth genus- b curve family, choosing a rational divisor of relative degree one and its pure projector as in Proposition 4.1. We use the operation algebra $\mathcal{A}_k(E)$ of Section 4 for an ordinary rational color space E . A *pure cubic entry* is the class of the projected small diagonal in $\mathcal{A}_3(E)$ with a specified triple of color inputs; products use disjoint curve slots.

We will fix an arbitrary color map $E \rightarrow W_b$, apply the surface theorem to a subspace with isotropic image, and use the following exterior-algebra lemma to annihilate every sufficiently long product. The final application of faithful polarization will then give a Chow identity with the same base-support bound.

Lemma 6.3 (An exterior ideal). *Let $F \subset H$ be vector spaces of dimensions $2r$ and D , and let $\Omega \in \wedge^2 F$ be nondegenerate. If $0 \leq t \leq r$, then the ideal generated by Ω^t in $\wedge H$ contains $\wedge^q H$ for every $q \geq D - r + t$.*

Proof. The linear symplectic Lefschetz decomposition gives surjectivity of

$$\Omega^t \wedge - : \bigwedge^{k-2t} F \longrightarrow \bigwedge^k F \quad (k \geq r + t).$$

Equivalently, decompose the exterior algebra into primitive Lefschetz strings; every term in those degrees is at least t positions up its string. Split $H = F \oplus F'$. A nonzero summand of $\wedge^q H$ has F' -degree at most $D - 2r$, hence F -degree at least $q - (D - 2r) \geq r + t$. It therefore lies in the stated ideal. $\square$

Lemma 6.4 (Arbitrary color products). *Let a smooth curve family of genus $b \geq 5$, with a section, be over a smooth irreducible quasi-projective base S satisfying $\dim S < (b - 1)^2 - 1$. Put*

$$t_b = 25(b - 1)^2 - 1.$$

Let E be an m -dimensional color space, and let e be an even integer with $e \geq m/2 - 2$ and $e \leq m/2$. Write $D_E = \binom{m}{3}$ and $d_e = \binom{e}{3}$. If $t_b \leq d_e/2$, every product of

$$q \geq D_E - d_e/2 + t_b$$

pure cubic entries whose colors lie in E vanishes modulo support in base codimension at least $b - 1$.

Proof. Fix a color test $z : E \rightarrow W_b$. The pullback to E of the symplectic form has a totally isotropic subspace of dimension at least $m/2$: its radical together with a Lagrangian in its symplectic quotient has that dimension. Choose an e -dimensional subspace E' of it. Its image under z is isotropic.

Choose an invertible skew matrix γ in a basis of E' . Grouping the ordered triples in Theorem 5.1, the coefficients of the resulting quadratic expression in cubic generators are a nonzero common scalar times the three-by-three minors of γ . Thus this expression is a nondegenerate bivector

$$\Omega \in \bigwedge^2 \left(\bigwedge^3 E' \right)$$

up to a nonzero scalar. Indeed, the third compound matrix is invertible, and its transpose is its negative because three is odd. In particular d_e is even. Theorem 5.1 says that the map from the exterior algebra on $\bigwedge^3 E$ to the odd cubic coefficients sends Ω^{t_b} to zero in the specified support quotient.

Lemma 6.3, with $F = \bigwedge^3 E'$ and $H = \bigwedge^3 E$, now kills every element of the asserted degree. This holds for every z ; although E' can depend on z , the degree threshold does not. Proposition 4.1 therefore proves the Chow statement. Its finite coefficient test also ensures that the resulting support still has codimension at least $b - 1$. $\square$

The smooth-family argument now gives vanishing of long products of pure cubic entries for arbitrary color inputs, modulo the stated base-support bound. The remaining task is to compare the compact candidate with these pure entries and control the errors along the boundary.

7 A cubic class on weighted compactifications

Lemma 6.4 gives vanishing of long products of projected small diagonals over smooth curve families, modulo the stated base support. To apply it on the entire moduli space, we construct a compact cubic operation whose restriction to a boundary chart differs from these pure diagonals by controlled terms. Weighted stable curves and reduction morphisms are due to Hassett [Has]. The smooth-core configurations below are related to the blowup compactifications of Fulton–MacPherson [FM] and their weighted relative versions [Rou]. For rational tails, Tavakol’s proposed comparison with powers of the universal curve [Tav] was established by Petersen [PetW, Proposition 2.11]. Here we also need control under cuts of nodal heavy cores. The correction at a rational bridge is essential: without it, a codimension-two exceptional class would remain on a three-light-point configuration space.

We distinguish three cubic objects. The formal class F_j^{st} belongs to the strata algebra and enters the definition of Y . For the empty heavy set, the corrected weighted cubic $F_{g,\emptyset}^0$ pulls back under weight reduction to $q(F_j^{\text{st}})$, with the corresponding label-forgetting pull. We also use its flagged versions $F_{h,B}^0$, called raw only to distinguish them from the auxiliary pure lifts $G_{h,B}$. Each $G_{h,B}$ is obtained from $F_{h,B}^0$ by controlled corrections: it agrees with the projected small diagonal over the smooth-core open, not globally on the compactification.

7.1 Weighted configurations and cut charts

Let (h, B) be stable heavy data: B is a finite ordered set and $2h - 2 + |B| > 0$. Write

$$X_{h,B}(k) = \overline{\mathcal{M}}_{h,(1^B, \epsilon^k)}, \quad 0 < k\epsilon \leq 1,$$

with $X_{h,B}(0) = \overline{\mathcal{M}}_{h,B}$. Here and below any positive rational weights of total at most one give the same moduli problem for the light labels. Lights can coincide with one another, but not with a

heavy marking. A rational component with one node and only light markings cannot be stable; a rational component with two nodes and positive light mass is stable. These observations also show the independence from the individual light weights.

The stacks $X_{h,B}(k)$ are smooth and proper, and reduction of weights and forgetting labels give the usual stabilization morphisms [Has, Theorem 2.1, Proposition 3.8, Theorems 4.1 and 4.3]. The one-light space is the universal curve over $\overline{\mathcal{M}}_{h,B}$: when its coarse position is a node or a heavy marking, a unique three-special-point rational component is inserted. Thus sections and the relative dualizing line define natural one-light classes.

Fix a stable graph Γ of the heavy type. At its vertex v let h_v be the genus and let B_v comprise the original heavy markings and the incident half-edges, all ordered. Put

$$\mathcal{B}_\Gamma = \prod_v \overline{\mathcal{M}}_{h_v, B_v}, \quad \mathcal{S}_\Gamma = \prod_v \mathcal{M}_{h_v, B_v}.$$

The *cut chart* with k lights is the disjoint union

$$X^\Gamma(k) = \coprod_{a: \{1, \dots, k\} \rightarrow V(\Gamma)} \prod_{v \in V(\Gamma)} X_{h_v, B_v}(a^{-1}(v)). \quad (7.1)$$

It has a proper gluing map to the weighted space of the original heavy type. Assignments are part of the indexing, so a class supported on the assignment of specified labels to a vertex means zero on the other components of (7.1).

Refining a graph gives further proper cut charts. These charts cover the preimage of the corresponding heavy boundary. Indeed, a node of the light-forgotten curve may have a chain above it before stabilization. Choose a node on that chain and cut it, replacing its branches by heavy flags. The resulting pieces remain weighted stable; gluing recovers the original curve. Several prescribed nodes can be cut independently. The map on heavy bases is finite, as for the ordinary gluing of stable curves.

Cutting and light forgetting commute. The heavy flag replacing a node branch retains exactly the stability contribution of that branch. Thus stabilizing on the separate pieces and then gluing gives the same curve as forgetting and stabilizing after gluing. This compatibility will permit operations on unconsumed light labels to pass through any later cut.

We use rational Chow classes operationally on these smooth stacks. For an ordinary color space U , the arity-indexed operation algebra is

$$\bigoplus_{k \geq 0} \left(A^*(X^\Gamma(k)) \otimes U^{\otimes k} \otimes \text{sgn}_k \right)_{\mathfrak{S}_k}. \quad (7.2)$$

The product pulls operations to disjoint sets of labels by forgetting and then multiplies them. Successive stabilization proves associativity; exchanging the two sets of labels gives the sign $(-1)^{k\ell}$. In particular, cubic and singleton operations are odd and pair operations are even. All expressions below are first identities of labeled classes and then identities in (7.2).

7.2 The corrected cubic

For three light labels define

$$F_{h,B}^0 = [1 = 2 = 3] - \frac{1}{2} \beta_{h,B} + \frac{1}{2} \sum_{p \in B} \delta_{p,123} \psi_p \in A^2(X_{h,B}(3)). \quad (7.3)$$

The coincidence class is the product of the divisors $[1 = 2]$ and $[1 = 3]$. The class $\beta_{h,B}$ is defined by taking each stable one-edge graph of the heavy type, subdividing its edge by a genus-zero vertex

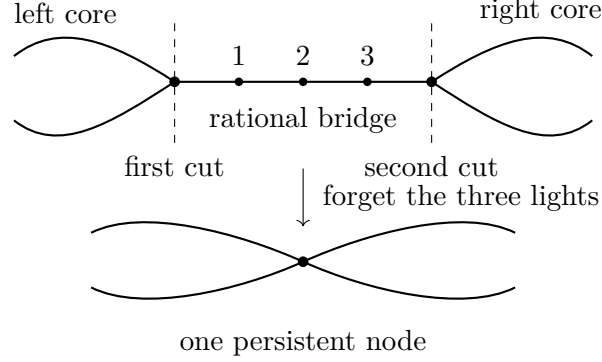


Figure 2: The two bounding nodes of a light-supported bridge. After forgetting the lights the bridge contracts. On a cut chart the two node conditions contribute two normal cotangent factors. The same calculation applies when the two core branches belong to one component. On the full exceptional surface, coincidence and the bridge class have integrals 1 and 2, respectively, giving $1 - \frac{1}{2} \cdot 2 = 0$.

carrying exactly the three lights, and summing the resulting gluing classes divided by their graph automorphism orders. The divisor $\delta_{p,123}$ has a tail containing exactly the heavy mark p and the three lights; ψ_p is the heavy cotangent class on the weighted space. Each term in (7.3) is symmetric in the three light labels.

Lemma 7.1. *Let $g \geq 2$, let the heavy set be empty, and pull the three-light operations to any larger number of light labels by forgetting. Under weight reduction from ordinary stable pointed curves, $[i = j]$ pulls back to $\bar{D}_{ij}$ and $\beta_{g,\emptyset}$ pulls back to the normalized two-edge bridge sum used to define β_j^{st} . Consequently the coefficient of the prescribed color word in the signed product of all $F_{g,\emptyset}^0$ pulls back to $q(Y)$, up to a nonzero rational scalar.*

Proof. In codimension one, the inverse image of coincidence consists of exactly the rational tails containing both labels. At the generic point of each tail, the difference of the two coarse positions is a node-smoothing parameter times a nonzero difference of coordinates on the tail. Every multiplicity is therefore one.

Consider next the two bounding nodes of a weighted bridge. Weight reduction retaining all labels contracts light-only tails, not a bridge with positive light mass. Thus both bounding nodes exist already in the source, and all the light labels assigned to the bridge originate between those cuts. The inverse image is supported on the corresponding ordinary strata of codimension at least two. At the generic two-edge stratum the three labels are distinct on the bridge and no component is contracted, so the pullback multiplicity is one. Further collisions add edges and have higher codimension. Normalized graph sums count these local cut choices once, including for a loop. This proves the supported codimension-two pullback assertion.

Forgetting commutes with reduction. Finally, extracting a fixed color word from signed averaging retains exactly the permutations preserving that word, with their signs. Depending on the averaging convention there is a positive factorial denominator, but no zero scalar. $\square$

7.3 The local three-position calculation

Over $\mathcal{S}_\Gamma$ let C_v denote the smooth core curve at vertex v . For a fixed assignment of three labels, there is a coarse position map

$$c : X^\Gamma(3)|_{\mathcal{S}_\Gamma} \longrightarrow C_{a(1)} \times_{\mathcal{S}_\Gamma} C_{a(2)} \times_{\mathcal{S}_\Gamma} C_{a(3)}. \quad (7.4)$$

All products in this subsection are over the indicated heavy base.

Lemma 7.2. *The map (7.4) is the iterated blowup obtained, at each flag p , by first blowing up the locus where all three assigned positions equal p , when present, and then the strict transforms of the loci where two assigned positions equal p . The latter centers are disjoint after the first blowup.*

Proof. Extra components above a smooth pointed core are chains attached at its flags. A branch with no heavy endpoint and only light markings would have a terminal light-only tail and is impossible. Every level of a chain carries positive light mass.

Choose a local parameter at a flag and write x_i for the corresponding coarse coordinate of label i . The zero divisor of x_i on the configuration space is the sum, with multiplicity one, of the cuts moving that label onto the expanded chain. The multiplicity follows from a one-node smoothing $xy = t$. At a fixed point of the configuration space these cuts form an initial segment of the ordered chain. Hence the products of smoothing parameters occurring in the different x_i are ordered by divisibility. In particular, each relevant pair or triple ideal is principal after pullback. This gives a morphism to the first blowup and then to the successive pair blowups; after the first step the pulled pair ideal is its strict transform ideal times the exceptional line.

The blowup ratios recover the expanded configuration. On a level put its outward endpoint at infinity and its inward endpoint at zero. The leading ratios of coarse coordinate differences are the ratios of the light coordinates on that level, modulo common scaling. Labels on separate levels give the limiting ratios zero or infinity. Forgetting to pairs gives all these ratios compatibly, so the ratios determine the levels and their configurations. Automorphisms fixing the core and the marked lights are trivial. The resulting representable map to the iterated blowup is therefore quasi-finite. It is also proper and birational. The target is smooth, and the source is smooth by weighted deformation theory; working on an etale chart of the core data, a proper birational quasi-finite map to this normal target is an isomorphism [Stacks, Tags 02LS and 0AB1].

In the first exceptional plane, the strict pair centers meet at the three distinct coordinate points, proving their disjointness. The constructions at distinct flags have disjoint centers. $\square$

Denote the total first exceptional divisor at a triple flag locus by $E_{p,123}$. Denote pair exceptional divisors, preferably pulled back from the corresponding two-light spaces, by $E_{p,ij}$. The blowup formula for Chow groups [Ful, Proposition 6.7(e)] shows that a codimension-two class on (7.4) has, in addition to its main pullback, terms of the following forms:

$$E_{p,ij} A_{p,ij}, \quad E_{p,123} D_p, \quad a_p E_{p,123}^2. \quad (7.5)$$

Here $A_{p,ij}$ is a divisor class on the remaining curve over the base, D_p is a base divisor class, and a_p is a scalar on each connected component of the base. For clarity, if $E_{p,ij}^{\text{str}}$ is the strict pair exceptional divisor and $E_{p,ij}^{\text{pull}}$ is pulled back from two labels, then

$$E_{p,ij}^{\text{pull}} = E_{p,ij}^{\text{str}} + E_{p,123}.$$

Multiplying the difference by a divisor on the remaining curve restricts that divisor, on $E_{p,123}$, to the flag section. It is therefore a pulled base divisor. Replacing the strict version by the pulled

version changes only the triple-exceptional-times-base-divisor term, and introduces no new scalar multiple of $E_{p,123}^2$. Formula (7.5) includes assignments to different vertices, in which some of the centers are simply absent.

When a raw class F^0 is pulled from an ancestor chart, its main component is the small diagonal if all three labels are assigned to one relevant descendant vertex, and is zero otherwise. Away from flags this is immediate. The coincidence, bridge, and heavy-tail corrections not seen there project to loci where all three coarse positions are flags, of codimension three; they cannot alter a main codimension-two class.

For the same pulled raw class, the divisor coefficients $A_{p,ij}$ in (7.5) are supported on flag sections of the remaining curve, up to base divisor terms. To see this, remove all flags from that remaining curve. Neither the raw class nor its main diagonal pullback meets the pair exceptional locus over this open. Intersecting with the exceptional divisor and pushing to its center recovers its coefficient, with the usual negative self-intersection sign for a codimension-two blowup center. The coefficient is therefore zero on this open. Chow localization on the curve over the base gives the assertion.

Lemma 7.3. *For the pullback of a raw cubic class F^0 from an ancestor chart, the coefficient a_p in (7.5) is zero at every triple flag locus.*

Proof. Work first over a field point of the smooth heavy base. The full fiber over three positions at one flag is the weighted chain space with two heavy endpoints and three lights, the Losev–Manin surface [LM, Theorems 2.2 and 2.6.3], [Has, §6.4]. By Lemma 7.2, it is the first exceptional $\mathbb{P}^2$ blown up at its three pair-collision points. The total class $E_{p,123}$ restricts to the pullback of $\mathcal{O}_{\mathbb{P}^2}(-1)$; hence

$$\int E_{p,123}^2 = 1. \quad (7.6)$$

Main pullbacks, pair terms multiplied by a remaining-position divisor, and base terms have integral zero on this fiber. Thus this integral detects exactly a_p .

The coincidence term gives one: its locus is the single transverse configuration in which the three lights coincide away from the endpoints. We distinguish new and inherited flags relative to the ancestor supplying the raw cubic. Suppose first that p is a new cut flag. There is one pair of bounding nodes isolating the chain carrying the three lights. This remains true on the boundary of the chain space. Indeed, a subcurve contracted upon forgetting the lights has two endpoint flags and no persistent heavy marking; every component has at least two incident branches before counting lights, so the component tree must be a chain, with positive light mass on every level. Its extreme bounding nodes are uniquely determined.

With these cuts distinguished, as in Figure 2, the bridge locus is a regular two-node condition in a versal deformation chart. Its two normal lines are the tensor products of the corresponding branch tangent lines. The core factors are constant on our fiber, so its self-intersection restricts to

$$(-\psi_0)(-\psi_\infty) = \psi_0\psi_\infty. \quad (7.7)$$

The normalized bridge sum counts this local branch once. This argument also treats a self-node: one distinguishes the two branches before applying the normal formula, and the graph automorphism factor removes the duplicate ordering.

Reduction from $\overline{\mathcal{M}}_{0,5}$ to the chain space preserves both heavy cotangent lines. A rational component carrying a heavy mark cannot be contracted when the other three weights are decreased but remain positive. The genus-zero string recurrence gives

$$\int_{\overline{\mathcal{M}}_{0,5}} \psi_0\psi_\infty = \int_{\overline{\mathcal{M}}_{0,4}} \psi_0 + \int_{\overline{\mathcal{M}}_{0,4}} \psi_\infty = 2. \quad (7.8)$$

The last equality follows by forgetting once more to $\overline{\mathcal{M}}_{0,3}$. Equivalently, (7.8) is the degree-two case of the genus-zero cotangent integral formula. The bridge contribution to F^0 is therefore $-\frac{1}{2} \cdot 2 = -1$, canceling coincidence.

If instead p is an inherited heavy flag, the relevant term is the unique outer tail with that heavy marking and all three lights. Its normal factor is $-\psi_0$ on the chain and the inserted ψ_p is ψ_∞ . Thus $\delta_{p,123}\psi_p$ has integral -2 , and its coefficient $+\frac{1}{2}$ again cancels coincidence. No other bridge or heavy-tail type meets this fiber: after the lights are forgotten its contracted locus is this specified node or this specified heavy mark.

We have shown that the scalar coefficient in (7.5) is zero on every connected smooth-core base component. This proves the result over that base. $\square$

7.4 Controlled pure lifts

For a smooth heavy core $\pi : C \rightarrow S$, choose a degree-one rational class

$$e_C = \begin{cases} [p], & \text{if a heavy flag } p \text{ has been chosen,} \\ c_1(\omega_{C/S})/(2h-2), & \text{if } B = \emptyset. \end{cases}$$

The second case occurs only for $h \geq 2$. As in the curve tensor construction, put

$$e'_C = e_C - \frac{1}{2}\pi^*\pi_*(e_C^2), \quad \eta_C = [\Delta_C] - e'_{C,1} - e'_{C,2}. \quad (7.9)$$

Write Δ_C^{pure} for the small diagonal projected by η_C in each of its three slots. Expanding the three projectors shows that

$$[\Delta_{123}] - \Delta_C^{\text{pure}} \quad \text{is a sum of pair-singleton products,} \quad (7.10)$$

with singleton factors 1_C or e'_C .

No numerical equivalence is used in this expansion: it is the decomposition of the diagonal correspondence into the pure projector and the two hyperbolic projectors.

Proposition 7.4 (Controlled pure lifts). *For every stable heavy type (h, B) one can choose a symmetric class $G_{h,B} \in A^2(X_{h,B}(3))$ with the following properties.*

- (i) *Over the smooth heavy base, $G_{h,B}$ is exactly the pullback of Δ_C^{pure} by the coarse-position map.*
- (ii) *Pull a raw cubic F^0 , or one of these chosen pure lifts G , from any ancestor chart to a compact descendant cut chart. Its operation is a sum of the chosen pure lifts at the relevant descendant vertices, with all three labels assigned to that vertex, together with pair-times-singleton operations and a residual cubic operation.*
- (iii) *The singleton operations in (ii), and all their further pullbacks through cuts and light forgetting, are controlled as follows. Over fixed smooth core data their restrictions lie componentwise in the span of the unit, the canonical class, and the flag points. On any smooth test family the same statement holds modulo classes supported over positive base codimension.*
- (iv) *On the current smooth-core open the residual operation in (ii) is a finite sum of classes containing a pulled base divisor factor. The base divisor may come from the current heavy base or from a compact ancestor base. Consequently its pullback to every smooth test family is zero operationally after removing a positive-codimension subset of that test base.*

All expansions preserve the color multiset. They may always use one and the same chosen G and centering at a given current vertex, regardless of which ancestor supplied the operation.

Proof. First work on the smooth heavy base of the class itself. The blowup decomposition (7.5) and Lemma 7.3 express F^0 as the pullback of the small diagonal, plus pair-exceptional-times-singleton terms and triple-exceptional-times-base-divisor terms. Subtract these terms, and also subtract (7.10).

These subtractions have compact extensions with the required structure. A pair exceptional divisor extends as the heavy-flag tail boundary on the two-light space; a triple exceptional divisor extends as its three-light counterpart. Divisor coefficients on the heavy base extend by Chow localization. Pair coefficients may also be extended by Chow localization on the smooth compact two-light stack. The singleton factors have specific extensions: use units, the heavy sections, the relative dualizing line on the one-light universal curve, and base classes. In particular we do not choose an arbitrary compact extension of the pure diagonal. Define G by making these structured subtractions from F^0 , and average over the three labels. This gives (i).

Now pull to a deeper cut chart. The raw F^0 has the decomposition just proved there. Its main component is the sum of the small diagonals of the relevant descendant vertices. Replace each by the corresponding chosen G plus its structured corrections. Inherited pair-times-singleton subtractions retain that form: forgetting commutes with cutting, so the pair coefficient remains a two-label operation and the singleton remains a one-label operation. The former is allowed to be arbitrary. For the latter, the normalization formula for the dualizing line gives the canonical class plus the new branch points on each component. A section either lies on the relevant component or has zero restriction there; base terms remain base terms. This proves the asserted span and its stability under further cuts.

The only other corrections are exceptional divisors multiplied by base divisor classes. On pulling an inherited correction, the exceptional factor need not have a special form, but its base divisor factor remains pulled from the ancestor heavy base. The same description applies to the new corrections on the current chart. Thus the residual over its smooth-core open is of the form

$$\sum_j H_j \cdot \rho_j^* D_j, \quad (7.11)$$

where H_j is a one-codimensional three-label operation and D_j is a divisor class on a current or ancestor heavy base, with the natural induced map ρ_j . Extending this decomposition to the compact chart can also leave a difference supported over its heavy boundary, by Chow localization. Include that difference in the residual operation. Its restriction to the smooth-core open is zero, so the residual there still has precisely the form (7.11).

To verify the last assertion after an arbitrary smooth test base change S , represent each pulled rational divisor class by the first Chern class of a rational line bundle. A rational trivialization makes it zero off a divisor of S . After removing the finite union of these divisors, every summand of (7.11) is zero as an operational class. This remains true if the image of S lies inside a divisor on an ancestor base: it is the pulled line bundle on S , not a presumed transverse pullback of its original supporting cycle, that is used. The base corrections in e'_G are covered by the same argument.

Finally, if a previous operation used a different centering or a different inherited presentation, expand it through its raw F^0 and the structured subtractions defining that presentation. The preceding argument again leaves the same allowed pair and singleton terms, and a residual of the stated form on the smooth-core open. We may therefore re-expand to the fixed choice of G at every current vertex. Assignment conditions can be imposed separately on the pair and singleton factors by making their operations zero on the other components of the disjoint union (7.1). All steps occur before inserting colors, and so preserve their multiset. $\square$

Remark 7.5. The conclusion in Proposition 7.4(iv) is stronger than vanishing on individual geometric fibers. It gives operational vanishing on an open of every smooth test family, including families mapping into a special locus of the heavy base. This is the form needed when the remaining operation is multiplied by an arbitrary cycle and Chow localization is used to lower its base support. Likewise, on the smooth-core open the equality in (i) is an equality of relative Chow correspondences. Smooth proper base change for their pull, intersection, and push operations identifies it with the centered tensor construction on any such test family.

8 Support induction and Chow vanishing

We now combine the arbitrary-color bound of Lemma 6.4 with the compact lifts of Proposition 7.4. We will make successive parts of the complete product of three-color entries vanish over dense opens of its base support, until no support remains. Throughout this section,

$$g = 10^{60}, \quad D_* = \binom{g}{3}, \quad n = 3D_*, \quad U_0 = \mathbb{Q}^g.$$

The basis of U_0 is the set of colors. An *entry* is one factor with three specified color inputs. Initially there is one entry for each increasing triple of distinct basis colors. Products and their factorizations are taken in the sign-coinvariant operation algebra of the weighted spaces from the preceding section.

The graph charts, their compact heavy bases, and their smooth-core opens will be denoted by

$$X^\Gamma(k), \quad \mathcal{B}_\Gamma, \quad \mathcal{S}_\Gamma \subset \mathcal{B}_\Gamma,$$

respectively. Every graph in this section has unmarked type $(g, 0)$; its vertex flags are heavy marks. An operation supported on a vertex is zero on assignment components on which its active labels are not all at that vertex.

8.1 Advancing support with an arbitrary multiplier

Boundary-support methods also appear in tautological vanishing results such as [GV, Theorem 1.1], where support is measured by rational components. Here we lower the dimension of the image in the heavy base, using operational vanishing over smooth test families. We first make this induction precise. A term in it consists of active operations, pulled to disjoint sets of light labels, acting on a cycle on the full labeled chart. The cycle need not factor over the labels or over the vertices.

Lemma 8.1 (Support advancement). *Let a cycle multiplier on $X^\Gamma(n)$ have base image contained in an irreducible closed subspace $Z \subset \mathcal{B}_\Gamma$. After passing to a refinement if necessary, suppose that its generic base image lies in $\mathcal{S}_\Gamma$. Let L be a factor involving a specified subset of the active entries.*

Suppose that the pull of L vanishes operationally over a dense open of a smooth test family dominating $Z \cap \mathcal{S}_\Gamma$. Then its action can be written as a finite sum of cycles with strictly smaller base-image dimension. Only the entries in L are consumed.

More precisely, take a smooth test base S , proper and generically finite over $Z \cap \mathcal{S}_\Gamma$. If the vanishing holds off base codimension at least a , the terms remaining over $\mathcal{S}_\Gamma$ have base-image dimension at most $\dim Z - a$. The other terms lie over $Z \setminus \mathcal{S}_\Gamma$ and require a strict graph refinement. If one first replaces S by the total space of one of its curve families to obtain a section, the corresponding bound is $\dim Z - a + 1$.

Proof. The proper cut-chart maps cover the preimages of boundary strata. Consequently a cycle whose generic heavy curve has a finer dual graph can be lifted to a chart of that graph. These lifts do not increase the dimension of its base image: heavy gluing is finite.

We use the following elementary property of rational Chow groups. If $T' \rightarrow T$ is proper and surjective, every integral cycle on T has, after multiplication by a nonzero rational number, a proper lift generically finite over it. Indeed, choose a closed point of the fiber over its generic point, take its finite residue-field extension, and close up that point. The pushforward is the original cycle multiplied by the generic degree. The same property holds for the finite-type Deligne–Mumford charts here with rational coefficients [BSch, Proposition B.19].

For the test base, take a finite level cover of the smooth heavy-curve stacks and resolve the appropriate component over $Z \cap \mathcal{S}_\Gamma$ [Wlo, Theorems 1.0.2 and 1.0.3]. This gives a smooth quasi-projective scheme proper and generically finite over that space. A full level of order at least three removes automorphisms in positive genus [DM, Sections 5.3–5.4 and 5.14], [DR, IV.2.7–2.9 and IV.3.1–3.2]; the genus-one vertices here have flags, and smooth genus-zero factors are already schemes. Lift the cycle multiplier properly to the resulting family by the preceding argument.

Let $A \subset S$ have codimension at least a , and suppose that L is zero operationally over $S \setminus A$. Its action on the lifted multiplier restricts to zero there. Chow localization expresses the result as a pushforward of cycles lying over A . Projection formula and proper pushforward return this equality to the original family. Because $\dim S = \dim Z$, these cycles have base-image dimension at most $\dim Z - a$. Closing up inside the inverse image of Z can introduce additional cycles only over $Z \setminus \mathcal{S}_\Gamma$, a proper closed subset of Z . Refining their generic dual graphs places them on the required cut charts.

To obtain a section of a smooth curve family $C \rightarrow S$, pull that family back along $C \rightarrow S$, where it has its diagonal section. The new test base is smooth and has dimension $\dim Z + 1$. Although this base change is not generically finite, the cycle multiplier still admits a proper lift generically finite over itself: apply the same generic-point argument to its pullback. A support of codimension at least a in the new base has image dimension at most $\dim Z + 1 - a$, proving the last assertion.

These operations concern the chosen factor L . All remaining operations are carried along by pullback and projection formula and can be applied afterwards to the resulting cycle multipliers. $\square$

The operational hypothesis in this lemma is essential. In its applications below, it follows from identities over smooth test families, rather than from identities separately on geometric fibers. For a pure operation, Proposition 7.4 identifies its restriction with a pullback from a smooth curve power. Vanishing of its Chow class on that power is therefore operational vanishing and can be pulled further to any cycle multiplier. The formation of the projector and the projected diagonal commutes with these base changes: the relative diagonals and the integrations along smooth proper curve powers have the requisite flat or refined-Gysin base-change identities. For residual operations, the explicit base-divisor factors in Proposition 7.4 give the required open vanishing. The same is true of the controlled singleton relations below.

There is also no restriction on the multiplier hidden in our use of sign coinvariants. In characteristic zero, signed averaging identifies coinvariants with the corresponding signed invariant subspace. A factorization $L * H$ of operations becomes a sum over shuffles of products of the separately averaged L and H , pulled to disjoint slot sets. In each summand, apply L first to the multiplier. The factor H remains an averaged operation on its own active slots; it is pulled equivariantly through later cuts before it acts. This proves that the lemma applies repeatedly, even after the multiplier has become an arbitrary cycle produced by an earlier step. More explicitly, every later factorization of H is first an equality of its signed-average representatives on its own active slots. Pull that equality to the full chart and only then cap with the multiplier. No permutation is applied to the multiplier,

and no invariance of the multiplier is required. Restriction to an assignment component or pullback through a further cut preserves this equality.

Initially the heavy base has dimension $3g - 3$. Every advancement strictly lowers the base-image dimension. Thus on any branch of the finite expansions there can be at most

$$3g - 2 \tag{8.1}$$

advancements before the support is empty.

8.2 The elementary vanishing tests

On a current chart, Proposition 7.4 expands an ancestor entry into pure entries at current vertices, products of a pair operation with a controlled singleton, and residual cubic operations. Call the pair-times-singleton terms *noise entries*. Perform these expansions one entry at a time. If a residual summand is chosen, apply Lemma 8.1 to it immediately, at the cost of that one entry. We therefore never retain residual entries while counting the pure and noise entries.

The pure entries at any one vertex always use the same chosen symmetric class $G_{h,B}$. This is important both for the finite coefficient bound and for the common-color test below.

Lemma 8.2 (Noise and small-genus tests). *Set*

$$R_\Gamma = 2|V(\Gamma)| + 2|E(\Gamma)|.$$

If $R_\Gamma + 1$ noise entries have their singleton at the same color, their product advances support while consuming those entries. If a genus- h vertex has more than $\binom{2h}{3}$ pure entries, the product of any $\binom{2h}{3} + 1$ of them advances support.

Proof. Over smooth generic core data, the one-light space at a vertex is its curve. The permitted singleton classes lie in the span of its unit, canonical class, and flag points. Over all vertices this gives a space of dimension at most

$$\sum_v (2 + |B_v|) = 2|V| + 2|E| = R_\Gamma.$$

A one-slot operation is odd in the sign-coinvariant algebra. At a fixed color, a product of $R_\Gamma + 1$ such elements is zero by multilinearity and alternation. The accompanying pair operations have even arity and can be grouped separately. This proves the first vanishing, and it holds on an open of each smooth test family by the controlled singleton descriptions and Chow localization.

For the second assertion apply Proposition 4.1. Every tested pure entry is a value of the same odd-coefficient form $\varphi \in \mathfrak{B} \otimes \bigwedge^3 W_h^\vee$. Its values belong to the span of at most $\binom{2h}{3}$ odd coefficients. A longer product vanishes by supercommutativity. The faithful same-map tensor test proves vanishing of the operation modulo positive base support. Only finitely many coefficient tests are needed in a fixed arity, so their supports can be combined without losing positive codimension. Apply Lemma 8.1. $\square$

8.3 A repeated-color test at a flagged vertex

The surface test of Lemma 6.4 is needed only while one vertex has almost all the genus. On every other chart a less expensive advancement follows from the flag-centered curve algebra.

Lemma 8.3 (Repeated-color advancement). *Suppose that a vertex of genus h has a flag and that $k > \binom{h+1}{2}$ pure entries contain one common color i . Their product is a finite linear combination of products $L * H$ with the following properties:*

- (i) L involves at most $h + 2$ entries and vanishes modulo positive base support;
 - (ii) H is still a product of individual pure entries at that vertex;
 - (iii) before removing the entries in L , each term has the same color multiset as the original product.
- Consequently each term admits a support advancement consuming at most $h + 2$ entries.

Proof. A symmetric cubic class defines an alternating three-color operation. Put the common color first in each entry. The product, as a function of the other $2k$ inputs, factors through

$$\bigwedge^k \left(\bigwedge^2 U_0 \right).$$

This holds already on the compact chart: alternation inside a pair comes from cubic symmetry, and interchange of two cubic entries has sign minus because their arity is odd.

Let $p \in \mathbb{Q}[\mathfrak{S}_{2k}]$ be the corresponding projector on the input slots. The exterior functor displayed above is zero on a vector space of dimension $h + 1$. By the Schur–Weyl decomposition and its row cutoff [EGH+, Theorems 5.12.2, 5.18.4, and 5.22.1], p lies in the two-sided ideal generated by the alternator on $h + 2$ slots: the quotient of the group algebra by that ideal acts faithfully on tensor powers in dimension $h + 1$. Indeed, its simple blocks are indexed by partitions; precisely those with at most $h + 1$ rows survive in that tensor power, and the $(h + 2)$ -alternator generates all the other blocks. Thus p is a finite sum of operators of the form

$$c \sigma \operatorname{Alt}_{h+2} \tau, \quad c \in \mathbb{Q}, \quad \sigma, \tau \in \mathfrak{S}_{2k}.$$

In one such term, first permute the color inputs by τ , and interpret σ as their assignment to the pairs of entries. Only entries meeting the $h + 2$ alternated positions vary during the alternation. There are at most $h + 2$ of them. Group their alternated product into L , leaving the other entries in H . This proves the asserted factorization and the color-multiset statement.

It remains to prove the vanishing of L . Work on a smooth test base modulo positive base support and center the curve projector at the chosen flag section. By Lemma 4.4, its class e has

$$\int e = 1, \quad \int e^2 = 0.$$

Evaluation at the section is pairing with e , is multiplicative, and kills the pure odd space W_h . Hence for standard odd vectors u, v , their product in the coefficient Frobenius algebra has the form

$$uv = \omega(u, v)e + (uv)_{\text{pure}}. \tag{8.2}$$

There is no unit component, since its coefficient is the value of uv at the section.

Test the colors by $z : U_0 \rightarrow W_h$, and write $a = z(i)$. Each entry becomes

$$T_a(u, v) = \varphi(a, u, v),$$

an alternating two-form with odd coefficients. Contracting a variable slot between two such entries gives zero. To see this, the full Frobenius contraction is, up to its fixed Koszul sign,

$$\int (au)(av) = 0,$$

because a is odd and $a^2 = 0$. The removed hyperbolic contraction in (8.2) is a multiple of $\int e^2 = 0$; orthogonality eliminates its mixed contractions. The remaining symplectic contraction of the pure products is exactly the stated contraction of the two cubic tensors.

An internal contraction of the two variable slots in a single T_a is one and the same odd scalar, say $\ell(a)$, up to the orientation sign. Two internal contractions in distinct entries therefore give zero, because $\ell(a)^2 = 0$.

After alternation, L is an ordinary alternating $(h+2)$ -form on W_h , with coefficients in the supercommutative algebra. Every summand of its double symplectic contraction is zero: if one contraction joins distinct entries, perform that external contraction first; otherwise the two contractions are internal in distinct entries and give the square just considered. All other variables can be held arbitrary in the external identity, so this reasoning also covers chains of contractions.

For $h \geq 2$, double contraction

$$\bigwedge^{h+2} W_h^\vee \longrightarrow \bigwedge^{h-2} W_h^\vee$$

is injective by symplectic Lefschetz decomposition. For $h = 0, 1$, the source alternation is already zero. Thus every color test of L vanishes. Proposition 4.1 and Lemma 8.1 finish the proof. $\square$

8.4 The finite color budget

Set

$$K = 10^5, \quad m_0 = 2000g^{2/3}, \quad C_K = \binom{2K}{3}. \quad (8.3)$$

These are integers. Reserve $4K + 20$ disjoint sets of m_0 basis colors each, and let S_0 be all the remaining colors. This is possible because

$$\frac{(4K + 20)m_0}{g} = 8.0004 \cdot 10^{-12} < 0.1; \quad |S_0| > 0.9g.$$

For a current graph put

$$b = \max_v h_v, \quad l = g - b.$$

Under refinement the maximum vertex genus cannot increase, so l cannot decrease. The number of edges cannot decrease either. Unmarked stability gives

$$|V| \leq 2g - 2, \quad |E| \leq 3g - 3, \quad R_\Gamma \leq 10g. \quad (8.4)$$

An advancement by a residual entry, by a noise test, by a small-genus test, or by Lemma 8.3 will be called *ordinary*. These are descriptions of the cost in entries, not additional geometric assumptions.

The two regimes below use different information about the remaining colors. While $l < K$, we remove whole entries without changing the three colors of any retained entry; the reserved blocks can therefore be counted by their original triples. We use the repeated-color rewrites only once $l \geq K$. They can change which colors occur together in an entry, but preserve the total multiset of color occurrences before the advancing factor is removed. In that regime we count occurrences in S_0 . Since l cannot decrease, we never need an original-triple count after making such a rewrite.

Small genus deficit: $l < K$. There is a unique vertex of genus b , since $b > g - K > g/2$. The total genus of all the others is at most l , and stability gives

$$|V| - 1 \leq 2l, \quad |E| \leq 3l, \quad R_\Gamma \leq 10K + 2. \quad (8.5)$$

For completeness, sum $2h_v - 2 + \text{val}(v) \geq 1$ over all vertices except the large one. The sum equals $2l - \text{val}(v_{\text{large}})$, proving the first inequality. The genus formula then gives the second.

After expanding the active entries, use a residual immediately. If a color occurs as a singleton in more than R_Γ noise entries, use Lemma 8.2. Otherwise there are at most $g(10K + 2)$ noise entries. If a small vertex has more than C_K pure entries, use the same lemma there. Otherwise the number of pure entries off the large vertex is at most $2KC_K$. Each ordinary advancement in this regime costs at most

$$M_K = \max\{10K + 3, C_K + 1\} = C_K + 1.$$

As specified above, retained entries in this regime still have their original triples, including those internal to each unused reserved block.

Suppose none of these ordinary tests applies. Choose a reserved color block E_0 that has not been used for a large-vertex test. Let $D_0 = \binom{m_0}{3}$. Among its original D_0 entirely internal entries, the number that are no longer active pure entries at the large vertex is at most

$$3gM_K + g(10K + 2) + 2KC_K < 10^{17}g < 0.01D_0. \quad (8.6)$$

Here (8.1) bounds all preceding ordinary losses. Earlier large-vertex tests used only entries internal to different, disjoint reserved blocks, so they removed none of these entries. The numerical inequalities follow from

$$C_K < 1.334 \cdot 10^{15}, \quad D_0 > 10^9 g^2, \quad g = 10^{60}.$$

Thus more than $0.99D_0$ internal entries remain pure at the large vertex.

Consume the product of all those entries. On a smooth test base, obtain a section if necessary as in Lemma 8.1. Its dimension is at most $3g - 2$; since $b > g - K$,

$$3g - 2 < (b - 1)^2 - 1.$$

We may therefore use Lemma 6.4. Choose an even integer e between $m_0/2 - 2$ and $m_0/2$. For $d_e = \binom{e}{3}$ and $t_b = 25(b - 1)^2 - 1$, the explicit bounds are

$$d_e > D_0/10, \quad t_b < 25g^2, \quad t_b < d_e/2,$$

and consequently

$$D_0 - d_e/2 + t_b < (0.95 + 2.5 \cdot 10^{-8})D_0 < 0.99D_0. \quad (8.7)$$

For the first bound, $m_0 \geq 128$ and

$$\frac{\binom{e}{3}}{\binom{m_0}{3}} \geq \left(\frac{1}{2} - \frac{4}{m_0}\right)^3 > \frac{1}{10}.$$

The selected product thus vanishes modulo base support of codimension at least $b - 1$. Applying Lemma 8.1, it either lowers the base-image dimension by at least $b - 2 > g - K - 2$, or gives a boundary remainder requiring a strict refinement. Call this an *expensive* advancement and mark E_0 as used.

There are enough fresh blocks for every such advancement. Those that require a refinement start with fewer than $3K$ edges by (8.5); along a branch there are at most $3K + 1$ of them. Every other expensive advancement lowers the base-image dimension by at least $g - K - 2$. There can be at most three such advancements with nonempty resulting support, because

$$4(g - K - 2) > 3g - 3.$$

A fourth can be terminal, making the support empty. In total at most $3K + 5 < 4K + 20$ blocks are needed. No expensive advancement consumes an occurrence of a color in S_0 .

Large genus deficit: $l \geq K$. Once this regime is reached it persists. We continue to expand ancestors using the same pure lift at each current vertex and peel residuals immediately. A noise test consumes at most $10g + 1$ entries. If no such test applies, at most $10g^2$ noise entries remain.

All ordinary advancements on the entire preceding path, including those in the small-deficit regime, cost at most $11g$ entries each:

$$C_K + 1 < 11g, \quad 10g + 1 < 11g, \quad h_v + 2 < 11g.$$

There have been at most $3g - 2$ advancements in total. Each consumed entry has three color occurrences. The cumulative ordinary loss of occurrences in S_0 is therefore less than $99g^2$. The unpeeled noise accounts for at most another $30g^2$. Expensive advancements used no S_0 colors. Hence the number of pure occurrences of colors from S_0 is at least

$$|S_0| \binom{g-1}{2} - 140g^2. \tag{8.8}$$

The allowance $140g^2$ is larger than the sum $129g^2$ of the two losses just obtained. The permutations used in Lemma 8.3 preserve the color multiset before its advancing factor is removed, so the same bound remains valid after any number of those rewrites.

Discard any identically zero product, in particular a product containing a pure entry with a repeated color. Suppose that every color $i \in S_0$, at every vertex v , belongs to at most $\binom{h_v+1}{2}$ pure entries. The total number of pure occurrences would then be at most

$$|S_0| \sum_v \binom{h_v+1}{2} \leq \frac{|S_0|}{2} (b+1) \sum_v h_v \leq \frac{|S_0|g(b+1)}{2}.$$

This contradicts (8.8). Indeed, the lower bound minus this upper bound is

$$\frac{|S_0|}{2} (g(l-4) + 2) - 140g^2 > (0.45(K-4) - 140)g^2 > 0.$$

There is therefore a color and vertex to which Lemma 8.3 applies. Every vertex in this regime has a flag: a connected graph without any flags at one of its vertices could only be the one-vertex, no-edge graph, for which $l = 0$. The lemma supplies another ordinary advancement, at the claimed cost.

8.5 Completion

Theorem 8.4. *The class $Y \in S_{g,n}$ constructed above satisfies*

$$q_{g,n}(Y) = 0 \quad \text{in } A^*(\overline{\mathcal{M}}_{g,n}; \mathbb{Q}).$$

Proof. Begin on $X_{g,\emptyset}(n)$ with the product of the cubic operations $F_{g,\emptyset}^0$, one for each increasing color triple, and with the fundamental cycle as multiplier. Apply Proposition 7.4 and the preceding induction. At every stage, each nonzero branch admits an advancement: use a residual or noise test when available, the small-vertex and reserved-block tests if $l < K$, and otherwise the repeated-color test. The counts show that all required active entries and unused blocks are available at the moment they are needed.

Each individual expansion and localization involves finitely many cycles. Each branch has length at most $3g - 2$, by (8.1). Thus the process ends with empty support and proves that the full product is zero in the sign-coinvariant Chow group of the weighted space.

The weighted-to-unweighted comparison in Lemma 7.1 identifies the pullback of the raw cubic with $q(F_J^{\text{st}})$. After signed averaging, the coefficient of the ordered color word $\mathbf{i}$ is a nonzero rational scalar times

$$q_{g,n} \left(\sum_{\tau: \tau \mathbf{i} = \mathbf{i}} \text{sgn}(\tau) \tau \left(\prod_J F_J^{\text{st}} \right) \right) = q_{g,n}(Y).$$

Pulling back the zero weighted class proves the theorem. $\square$

Proof of Theorem 1.1. Theorem 8.4 places Y in the Chow kernel. Proposition 3.1 supplies a linear map annihilating the specified Pixton span and taking a nonzero value on Y . Thus

$$Y \in \ker q_{g,n} \setminus \mathcal{P}_{g,n}.$$

The stable pair $g = 10^{60}$, $n = 3\binom{g}{3}$ is a counterexample to the proposed equality for all stable pairs. $\square$

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